

One-bit Single-Measurement Spectral Compressed Sensing via Hankel Matrix Factorization

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Abstract—This paper studies the problem of recovering a spectrally sparse signal from its one-bit single-measurement vector. Most previous methods depend on multiple measurement vectors, which face inherent challenges in real-time critical scenarios. Regarding the single-measurement vector, only a few on-grid approaches have been proposed; however, these techniques suffer from estimation errors due to grid mismatch. To facilitate gridless reconstruction under a one-bit single-measurement vector, a novel optimization model is devised by leveraging both the low-rank property of the lifting Hankel matrix and the sign consistency between the target signal and its estimate. Consequently, a one-bit spectral compressed sensing algorithm via Hankel matrix factorization, termed OSCAR, is proposed by incorporating a one-sided ℓ_p norm smoothing technique. Furthermore, an accelerated OSCAR algorithm, named Acc-OSCAR, is developed by integrating momentum techniques to expedite convergence. Theoretical results provide the convergence analysis for the proposed algorithms. Numerical results confirm the excellent performance of the proposed algorithms.

Index Terms—Spectral compressed sensing, one-bit, Hankel matrix completion, matrix factorization

I. INTRODUCTION

One-bit spectral compressed sensing aims to recover spectrally sparse signals from one-bit measurements. Owing to its hardware efficiency and energy-saving advantages in analog-to-digital conversion, one-bit spectral compressed sensing has attracted significant attention in applications ranging from automotive radar and underwater sonar to channel estimation [1], [2].

Let $\mathbf{x}^* = [x_0^*, x_1^*, \dots, x_{n-1}^*]^T \in \mathbb{C}^n$ denote the original single-measurement vector for a spectrally sparse signal,

whose elements are a superposition of r complex sinusoids

$$x_t^* = \sum_{a=1}^r d_a e^{(i2\pi f_a - \tau_a)t}, \quad (1)$$

where $t = 0, \dots, n-1$, $f_a \in [0, 1)$ is the a -th normalized frequency, d_a is the amplitude of the a -th sinusoid, and τ_a is the a -th damping factor. Let $\mathbf{y}$ be its one-bit sampling, defined as

$$\mathbf{y} = \text{csgn}(\mathbf{x}^*) = \sum_{t=1}^n \text{csgn}(x_t^*) \mathbf{e}_t, \quad (2)$$

where $\mathbf{e}_t$ is the canonical basis of $\mathbb{R}^n$, $\text{csgn}(\cdot)$ is defined as

$$\text{csgn}(x) = \text{sgn}(\text{Re}(x)) + j \text{sgn}(\text{Im}(x)), \text{ for } x \in \mathbb{C}, \quad (3)$$

$\text{Re}(\cdot)$ and $\text{Im}(\cdot)$ denote the real and imaginary parts of a complex scalar, respectively and $\text{sgn}(\cdot)$ is defined as

$$\text{sgn}(h) = \begin{cases} +1, & h \geq 0, \\ -1, & h < 0, \end{cases} \text{ for } h \in \mathbb{R}. \quad (4)$$

Here, we assume that n is odd as in [3]; otherwise, we can delete the last entry of $\mathbf{y}$ to make it odd.

By simplifying signals into binary representations, one-bit spectral compressed sensing significantly reduces data volume, thereby lowering sampling and storage costs. Meanwhile, replacing analog-to-digital converters with comparators minimizes hardware costs and power consumption. However, one-bit sampling inevitably results in the loss of both amplitude and phase information, thereby complicating the reconstruction of the original signal. While the exact amplitude of the desired signal cannot be recovered, it is still possible to reconstruct a scaled version of the signal from one-bit measurements, from which the underlying frequencies and damping factors can be estimated. This naturally raises the question: *how can we efficiently and accurately reconstruct a scaled version of the desired signal?*

To recover the scaled desired signal from one-bit measurements, several strategies have been proposed, including on-grid approaches based on one-bit compressed sensing [4], [5], [6], [7], one-bit atomic norm minimization methods [8], [9], [10], [11], and deep learning-based algorithms [11], [12], [13], [14]. However, most of the current algorithms focus on multiple measurement vectors, which demonstrate inherent limitations in real-time critical scenarios, particularly in high-mobility applications such as autonomous driving systems and unmanned combat platforms. Only a limited number of on-grid algorithms demonstrate applicability in single-measurement vector scenarios, but these methods suffer from performance degradation when subjected to grid mismatch conditions.

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This paper has supplementary downloadable material available at <http://ieeexplore.ieee.org>, provided by the author. The material includes a numerical verification of the KL property for Eq. (73), the impact of rank mis-specification and the smoothing parameter on performance, and the full proof of Lemma 2. This material is 592KB in size.

A. Contributions

This paper considers achieving gridless spectral compressed sensing under a one-bit single-measurement vector setting based on low-rank Hankel matrix factorization. The main contributions are as follows:

- 1) A non-convex optimization model is formulated by exploiting the low-rank characteristic of the lifting Hankel matrix, along with ensuring sign consistency between the original and estimated signals, as shown in Fig. 1. The low-rank Hankel matrix factorization is utilized to reduce the dimension of the variables, while the one-sided ℓ_p norm is employed to encourage sign consistency.
- 2) We develop a one-bit single-measurement spectral compressed sensing method via Hankel matrix factorization called OSCAR by utilizing a one-sided ℓ_p norm smoothing technique. Additionally, we present an accelerated variant, Acc-OSCAR, which incorporates momentum strategies to improve convergence performance. Furthermore, to boost computational efficiency, we implement a rapid gradient computation strategy for the proposed algorithms by employing the Fast Fourier Transform (FFT).
- 3) We conduct a theoretical analysis and provide experimental validation for the algorithm's convergence performance in single-measurement vector scenarios, thereby demonstrating the efficacy of the proposed OSCAR methods.

B. Literature Review

Standard spectral compressed sensing considers the reconstruction of spectrally sparse signals from unquantized measurements. There are two main kinds of gridless algorithms for standard spectral compressed sensing, which are the atomic norm minimization (ANM) method and the Hankel matrix completion method. The atomic norm minimization method first designs an atomic norm over a continuous dictionary, then establishes an atomic norm minimization model, and solves it through an equivalent semidefinite programming formulation. For instance, to relax the requirement on spectral separation, Yang and Xie proposed a reweighted atomic norm minimization method that enhanced spatial spectral estimation accuracy under single-measurement conditions [15]. Qian et al. developed a generalized atomic norm-based model, achieving precise recovery of missing data in multiple measurement vectors [16]. The Hankel matrix completion method exploits the low-rank property of the lifted Hankel matrices of the original signal, formulates a low-rank optimization model, and solves it via convex/nonconvex optimization algorithms, which effectively addresses spectral compressed sensing. For example, Chen and Chi proposed a structured matrix completion algorithm that achieves high-probability exact signal reconstruction from a single-measurement vector [17]. Cai et al. studied a projected gradient descent based on asymmetric Hankel factorization and achieved linear convergence [18]. Li et al. introduced a symmetric Hankel structure to the optimization model, which enables exact signal reconstruction

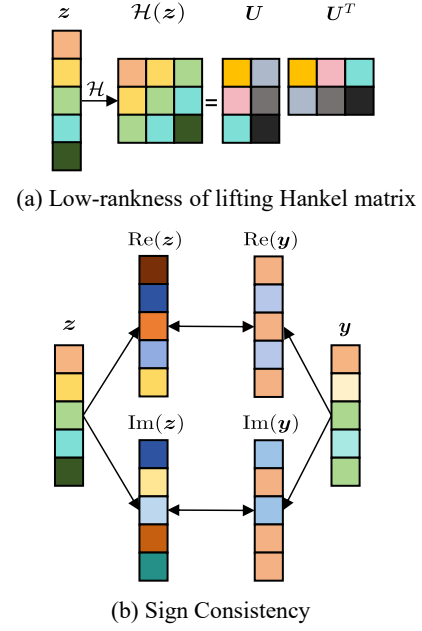


Fig. 1. Low rankness of lifting Hankel matrix and sign consistency.

with less storage and computational time under a single-measurement vector [3].

Building upon unquantized spectral compressed sensing and one-bit compressed sensing frameworks, researchers have begun to study one-bit spectral compressed sensing, which mainly includes:

1) One-bit subspace methods via covariance matrix estimation. These approaches first estimate the actual covariance matrix from one-bit data before applying subspace-based spectrum estimation. For instance, Huang and Liao proposed the one-bit multiple signal classification (MUSIC) algorithm [19], which employs a weighted combination of one-bit covariance matrices and identity matrices for covariance estimation, achieving comparable performance to unquantized MUSIC at low signal-to-noise ratio (SNR). Xiong et al. enhanced estimation accuracy through exchange matrix operations for covariance reconstruction under low SNR conditions [20]. However, these methods demonstrate significant performance gaps compared to unquantized subspace approaches at high SNR. Besides, they are limited to multiple measurement vectors, failing in the case of single-measurement vectors.

2) On-grid methods via one-bit compressed sensing. These techniques construct grid-based steering matrices and employ one-bit compressed sensing algorithms for spectral estimation. Stöckle et al. proposed a complex binary iterative hard-thresholding algorithm with low computational complexity [4]. Wang et al. developed a grid-based approach that incorporates basis update and backtracking strategies and refined threshold functions to address off-grid issues [5]. Chen et al. solved a maximum likelihood estimator through conjugate gradient and steepest descent methods [6]. Li et al. carried out dual-sparsity optimization with proximal alternating minimization for sign-flip correction [7]. Li et al. developed a nonzero-threshold quantization model and proposed a mixed-norm optimization, achieving automatic source-number and

amplitude estimation with improved performance in low-SNR regimes [21]. While current grid-based methods handle both single- and multi-measurement vectors, their estimation accuracy remains limited by grid mismatch errors.

3) One-bit atomic norm minimization approaches. These methods formulate convex optimization problems through sign consistency and atomic norm constraints, subsequently converted to semidefinite programming for scaled signal recovery. Wei et al. proposed a gridless one-bit DOA estimation approach via atomic norm denoising [22]. Tang et al. solved binary atomic norm minimization using accelerated proximal gradient [8]. Zheng and Yang exploited Hankel-Toeplitz structures to conduct reweighted atomic norm minimization for multiple measurement vectors [9]. Li et al. carried out log-determinant-based approximate atomic norm minimization with noise constraints [10]. Feng et al. combined maximum a posteriori estimation with joint basis-sparse signal optimization in an off-grid iterative reweighting algorithm [1]. Despite their high estimation accuracy, these convex optimization methods suffer from prohibitive computational complexity, which limits their practical implementation.

4) Deep learning-driven one-bit algorithms. Emerging deep learning approaches aim to enhance computational efficiency, though research remains in its infancy. Xiao et al. developed a deep unfolding model named DeepFPC for a single-measurement vector by integrating grid-based methods with sign consistency, improving mid-to-high SNR performance [11]. Ji et al. proposed a Transformer-enhanced subspace network for multiple measurement vectors, outperforming conventional subspace methods under both low-bit and one-bit quantization conditions [12]. Hu et al. proposed a learning-based sparse Bayesian framework that integrates unrolled neural networks to achieve high-precision off-grid estimation from a single-measurement vector [14]. However, these algorithms rely on large amounts of training data and usually require fixing the dimension of the input signals.

C. Organization and Notations

Organization. The paper is organized as follows. Section II provides the problem formulation by exploring the low-rankness of the lifting Hankel matrix and the sign consistency between the original signal and the estimated signal. Section III proposes the smooth approximation of the objective function. Section IV presents the OSCAR algorithm and its accelerated version Acc-OSCAR. Section V provides the performance guarantees for the proposed algorithm. Numerical simulations are given in Section VI, and conclusions are provided in Section VII.

Notations. For a natural number K , the symbol $[K]$ denotes $[K] = \{0, 1, \dots, K-1\}$. For a vector $\mathbf{x} \in \mathbb{C}^n$, let $\bar{\mathbf{x}}$, $\mathbf{x}^T$, and $\mathbf{x}^H$ denote its element-wise complex conjugate, transpose, and conjugate transpose, respectively. For a matrix $\mathbf{X} \in \mathbb{C}^{m \times n}$, let $\bar{\mathbf{X}}$, $\mathbf{X}^T$, $\mathbf{X}^H$ denote its element-wise complex conjugate, transpose, and conjugate transpose, respectively. Let $\mathbf{I}_s$ denote the identity matrix with dimension $s \times s$. The symbol $\text{vec}(\mathbf{X})$ returns an $mn \times 1$ vector in column-major order, and $\text{diag}(\mathbf{A}) \in \mathbb{C}^{mn \times mn}$ is a diagonal matrix formed by arranging the elements of $\mathbf{A} \in \mathbb{C}^{m \times n}$ in column-major order.

The inner product of two vectors is defined as $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^H \mathbf{y}$ for $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ and the inner product of two matrices is defined as $\langle \mathbf{X}, \mathbf{Y} \rangle = \text{tr}(\mathbf{X}^H \mathbf{Y}) = \text{vec}(\mathbf{X})^H \text{vec}(\mathbf{Y})$ for $\mathbf{X}, \mathbf{Y} \in \mathbb{C}^{m \times n}$. The largest entry of a matrix $\mathbf{X}$ is defined as $\|\mathbf{X}\|_\infty$. The ℓ_p norm of a vector $\mathbf{x}$ is termed as $\|\mathbf{x}\|_p$. The term $\|\cdot\|_{2,\infty}$ denotes the largest ℓ_2 -norm of its rows.

II. PROBLEM FORMULATION

This section formulates the optimization model by leveraging two properties of the estimated spectrally sparse signal. In particular, the model exploits the low-rankness of the lifting Hankel matrix of the estimated signal and the sign consistency between the original signal and the estimated signal as follows:

1) Low-rankness of lifting Hankel matrix. Previous work [18], [23] has shown that the lifting Hankel matrix $\mathcal{H}(\mathbf{x}^*) \in \mathbb{C}^{n_1 \times n_2}$ is of low rank, that is,

$$\text{rank}(\mathcal{H}(\mathbf{x}^*)) = r, \quad (5)$$

where $\mathcal{H}$ is the Hankel lifting operator that maps a vector $\mathbf{x} \in \mathbb{C}^n$ to a matrix $\mathcal{H}\mathbf{x} \in \mathbb{C}^{n_1 \times n_2}$ with $[\mathcal{H}\mathbf{x}]_{i,j} = x_{i+j}$, $i \in [n_1]$, $j \in [n_2]$ and $n_1 + n_2 - 1 = n$. Due to one-bit quantization, only the direction of the signal can be recovered. Here, the optimization variable $\mathbf{z} \in \mathbb{C}^n$ denotes an estimate of the underlying true signal $\mathbf{x}^*$ up to a positive scalar factor, i.e., $\mathbf{z} = \alpha \mathbf{x}^*$ with $\alpha > 0$, due to the inherent scale ambiguity of one-bit measurements. Thus, we require that the estimated signal $\mathbf{z}$ satisfies

$$\text{rank}(\mathcal{H}(\mathbf{z})) = r. \quad (6)$$

2) Sign consistency. According to the definition of one-bit sampling, we have

$$\begin{pmatrix} \text{Re}(\mathbf{y}) \odot \text{Re}(\mathbf{x}^*) \\ \text{Im}(\mathbf{y}) \odot \text{Im}(\mathbf{x}^*) \end{pmatrix} \succeq 0, \quad (7)$$

which leads to

$$\text{Re}(\bar{\mathbf{y}} \odot \mathbf{x}^*) = \text{Re}(\mathbf{y}) \odot \text{Re}(\mathbf{x}^*) + \text{Im}(\mathbf{y}) \odot \text{Im}(\mathbf{x}^*) \succeq 0, \quad (8)$$

where $\odot$ denotes the Hadamard product, $\mathbf{a} \succeq 0$ means $a_k \geq 0, k = 1, \dots, n$, for $\mathbf{a} \in \mathbb{R}^n$. Therefore, to enforce the sign consistency of the one-bit sampling $\mathbf{y}$ and the estimated signal $\mathbf{z}$, the number of negative elements in $\text{Re}(\bar{\mathbf{y}} \odot \mathbf{z})$ should be as small as possible. Inspired by the binary iterative hard thresholding (BIHT) approach in standard compressed sensing [24], we use the convex relaxation called one-sided ℓ_p -norm to achieve this

$$\left\| [\text{Re}(\bar{\mathbf{y}} \odot \mathbf{z})]_- \right\|_p^p, \quad (9)$$

where $[\mathbf{q}]_-$ keeps the negative elements in $\mathbf{q} \in \mathbb{R}^n$ and sets the nonnegative elements in $\mathbf{q}$ to be zero, and $p \in \mathbb{Z}^+$ is a positive integer. Like [24] and [4], we choose $p = 1$ and $p = 2$ in our algorithm, where $p = 1$ is suitable for sparse noise and $p = 2$ is suitable for dense noise.

Remark 1. An alternative choice of the objective function (9) is

$$\left\| \begin{bmatrix} \text{Re}(\mathbf{y}) \odot \text{Re}(\mathbf{z}) \\ \text{Im}(\mathbf{y}) \odot \text{Im}(\mathbf{z}) \end{bmatrix} \right\|_p^p. \quad (10)$$

But it makes the algorithm and expressions more complicated and shows no benefit according to the simulations (c.f. Figs. 3 and 4), so we choose to use (9) to enforce the sign consistency.

Notice that Eqs. (5) and (8) hold for any scale of $\mathbf{x}^*$. To eliminate this scaling effect, we suppose without loss of generality that $\|\mathcal{H}(\mathbf{x}^*)\|_F = 1$ and require the estimate $\mathbf{z}$ to satisfy

$$\|\mathcal{H}(\mathbf{z})\|_F^2 = 1 \quad (11)$$

for ease of analysis¹. Then the optimization problem becomes

$$\begin{aligned} \min_{\mathbf{z}} \quad & \left\| [\text{Re}(\bar{\mathbf{y}} \odot \mathbf{z})]_- \right\|_p^p \\ \text{s.t.} \quad & \text{rank}(\mathcal{H}(\mathbf{z})) = r, \|\mathcal{H}(\mathbf{z})\|_F^2 = 1. \end{aligned} \quad (12)$$

Given the complexity associated with the rank constraint (6), the problem presented is NP-hard. Given the fact that n is odd, we employ symmetric Hankel matrix factorization as an alternative to Eq. (6), that is,

$$\mathcal{H}(\mathbf{z}) = \mathbf{U}\mathbf{U}^T, \mathbf{U} \in \mathbb{C}^{s \times r}, \quad (13)$$

where $s = (n+1)/2$ is an integer.

To enforce the Hankel structure of $\mathbf{U}\mathbf{U}^T$, we introduce some important operators. We denote $\mathcal{H}^*$ as the adjoint of $\mathcal{H}$, which maps a matrix $\mathbf{M} \in \mathbb{C}^{s \times s}$ to a vector $\mathcal{H}^*(\mathbf{M}) = \{\sum_{i+j=a} \mathbf{M}_{i,j}\}_{a=0}^{n-1} \in \mathbb{C}^n$. Let $\mathcal{D}^2 = \mathcal{H}^* \mathcal{H}$ and $\mathcal{D}$ be a linear operator that maps a vector $\mathbf{q} \in \mathbb{C}^n$ to $\mathcal{D}\mathbf{q} \in \mathbb{C}^n$, where $[\mathcal{D}\mathbf{q}]_a = \sqrt{w_a} q_a$ and w_a is the length of the a -th skew-diagonal of an $s \times s$ matrix with $a \in [n]$. Define $\mathcal{G} = \mathcal{H}\mathcal{D}^{-1}$. Since $\mathcal{D}$ is invertible, we have $\mathcal{G}^* \mathcal{G} = \mathcal{I}$, where $\mathcal{I}$ denotes the identity operator. Then we can enforce the Hankel structure [3] as follows

$$(\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T) = 0, \quad (14)$$

where $\mathcal{I} - \mathcal{G}\mathcal{G}^*$ is a projection operator.

Since $\mathcal{G}^* \mathcal{G} = \mathcal{I}$ and $\mathcal{H} = \mathcal{G}\mathcal{D}$, we have $\mathbf{z} = \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)$. Therefore, we reformulate the optimization problem as

$$\begin{aligned} \min_{\mathbf{U}} \quad & \left\| [\text{Re}[\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)]]_- \right\|_p^p \\ \text{s.t.} \quad & (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T) = 0, \|\mathbf{U}\mathbf{U}^T\|_F^2 = 1. \end{aligned} \quad (15)$$

Notice that Problems (12) and (15) are equivalent: substituting the partial isometry $\mathcal{G} = \mathcal{H}\mathcal{D}^{-1}$ into (12) and reparameterizing the rank- r complex-symmetric matrix $\mathcal{H}(\mathbf{z})$ by the symmetric factorization $\mathbf{U}\mathbf{U}^T$ yields (15) directly.

We aim to find the best rank- r approximation that minimizes the negative sign error and satisfies the Hankel structure and the norm constraint. Following previous work [18], [3], we constrain the solution $\mathbf{U}$ in a convex set $\mathcal{C}$, which is defined as

$$\mathcal{C} = \left\{ \mathbf{U} \in \mathbb{C}^{s \times r} \mid \|\mathbf{U}\|_{2,\infty} \leq 2\sqrt{\frac{\mu r \sigma}{n}} \right\}, \quad (16)$$

where $\mu > 0$ is an incoherence constant and $\sigma = \sigma_1(\mathcal{H}(\mathbf{x}^*))$ denotes the maximum singular value of $\mathcal{H}(\mathbf{x}^*)$. The projection onto the set $\mathcal{C}$ encodes the known structural prior of the factor matrix as a convex constraint [18], [3]. This enforces the desired incoherence property derived from the spectral signal's

¹The right-hand side can be chosen as an arbitrary positive constant without influencing the estimation of frequencies and damping factors.

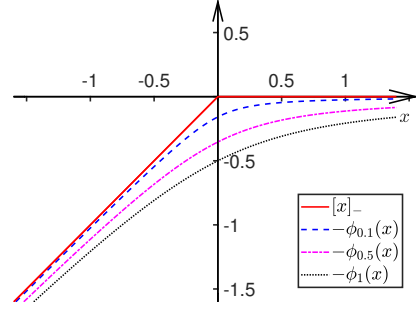


Fig. 2. Comparisons of $[x]_-$ and its smoothing function $-\phi_b(x)$ under different smoothing parameters b .

characteristics. The convexity and boundedness of this set are key to establishing a well-defined Lipschitz constant, enabling rigorous convergence guarantees for our algorithms.

Finally, we penalize the first constraint in the objective function, which is formulated as

$$\begin{aligned} \min_{\mathbf{U}} \quad & F_p(\mathbf{U}) \triangleq \frac{1}{p} \left\| [\text{Re}[\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)]]_- \right\|_p^p \\ & + \gamma \|(\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T)\|_F^2 \quad (17) \\ \text{s.t.} \quad & \mathbf{U} \in \mathcal{C}, \|\mathbf{U}\mathbf{U}^T\|_F^2 = 1. \end{aligned}$$

where $\gamma > 0$ is a tradeoff parameter. Problem (17) is a penalized relaxation of (15): the hard Hankel-structure constraint is replaced by the quadratic penalty $\gamma \|(\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T)\|_F^2$, so exact equivalence holds only as $\gamma \rightarrow \infty$. In practice, a sufficiently large γ makes the Hankel-structure violation numerically negligible.

Define

$$f_p(\mathbf{U}) = \frac{1}{p} \left\| [\text{Re}[\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)]]_- \right\|_p^p, \quad (18)$$

$$g(\mathbf{U}) = \|(\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T)\|_F^2. \quad (19)$$

Then we have

$$F_p(\mathbf{U}) = f_p(\mathbf{U}) + \gamma g(\mathbf{U}). \quad (20)$$

It should be noted that $[\cdot]_-$ is non-smooth, and both the objective function and the constraint are nonconvex. Therefore, Problem (17) is a nonsmooth and nonconvex problem. The absence of smoothness impedes the application of momentum acceleration techniques for algorithm design and makes it difficult to analyze the convergence of the corresponding algorithms. To address this issue, a smooth approximation of the objective function is introduced in Section III.

III. SMOOTH APPROXIMATION

In this section, we study the smooth approximation of Problem (17) to make it easier for the acceleration algorithm design and convergence analysis.

Note that $[q]_- = -\max(-q, 0) = -\text{ReLU}(-q)$, where ReLU is a classical activation function in deep learning. Motivated by the smoothing technique of ReLU [25], we propose to use a smooth function $-\phi_b(q)$ to replace the original one-sided negative function $[q]_-$, which is an element-wise function defined as

$$[\phi_b(q)]_i = \phi_b(q_i) = \frac{\sqrt{q_i^2 + b} - q_i}{2}, \quad (21)$$

where $b > 0$ is a smoothing parameter. When $b = 0$, $-\phi_b(\mathbf{q})$ degrades to $[\mathbf{q}]_-$. As shown in Fig. 2, $-\phi_b(\mathbf{q})$ approximates $[\mathbf{q}]_-$ from below, and the smaller b is, the closer $-\phi_b(\mathbf{q})$ is to $[\mathbf{q}]_-$.

By simple calculation, for any $q_i \in \mathbb{R}$, we obtain

$$\phi_b(q_i) > 0, \quad (22)$$

$$\phi'_b(q_i) = \frac{1}{2} \left(\frac{q_i}{\sqrt{q_i^2 + b}} - 1 \right) < 0, \quad (23)$$

$$\phi''_b(q_i) = \frac{b}{2(q_i^2 + b)^{\frac{3}{2}}} > 0, \quad (24)$$

where $\phi'_b(q)$ and $\phi''_b(q)$ denote the first-order gradient and second-order gradient of ϕ_b at $q \in \mathbb{R}$, respectively.

Therefore, we can change the optimization problem from the nonsmooth counterpart to the following smooth one

$$\begin{aligned} \min_{\mathbf{U}} \hat{F}_p(\mathbf{U}) &\triangleq \frac{1}{p} \left\| \phi_b(\text{Re}[\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)]) \right\|_p^p \\ &\quad + \gamma \left\| (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T) \right\|_F^2 \quad (25) \\ \text{s.t. } \mathbf{U} &\in \mathcal{C}, \left\| \mathbf{U}\mathbf{U}^T \right\|_F^2 = 1. \end{aligned}$$

Define

$$\hat{f}_p(\mathbf{U}) = \frac{1}{p} \left\| \phi_b(\text{Re}[\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)]) \right\|_p^p, \quad (26)$$

then we have

$$\hat{F}_p(\mathbf{U}) = \hat{f}_p(\mathbf{U}) + \gamma g(\mathbf{U}). \quad (27)$$

Compared with Problem (17), there are two main benefits of Problem (25): 1) The objective function of (25) is smooth; we can adopt momentum-based approaches to accelerate the algorithm if necessary; 2) While the convergence analysis is difficult for the nonconvex and nonsmooth objective function $F_p(\mathbf{U})$, it is relatively simple for the nonconvex but smooth objective function $\hat{F}_p(\mathbf{U})$.

In the subsequent sections, we will propose algorithms for Problem (25) in Section IV and give the related convergence analysis in Section V.

IV. ALGORITHM

This section proposes a one-bit spectral compressed sensing algorithm via Hankel matrix factorization (OSCAR) for problem (25) and its accelerated variant Acc-OSCAR. First of all, we present the initialization scheme. The initial point $\mathbf{U}^0$ is set as follows: Let the Takagi factorization of $\mathcal{H}(\mathbf{y})$ be

$$\mathcal{T}_r(\mathcal{H}(\mathbf{y})) = \mathbf{U}_y \mathbf{S}_y \mathbf{U}_y^T, \quad (28)$$

where $\mathbf{U}_y \in \mathbb{C}^{s \times r}$ is a unitary matrix and $\mathbf{S}_y \in \mathbb{C}^{r \times r}$ is a real nonnegative diagonal matrix. After normalizing the spectral initialization as $\mathbf{U}_y \mathbf{S}_y \mathbf{U}_y^T / \|\mathbf{S}_y\|_F$, the initial factor $\mathbf{U}^0$ is constructed as follows:

$$\mathbf{U}^0 = \mathcal{P}_C(\mathbf{U}_y \mathbf{S}_y^{\frac{1}{2}} / \|\mathbf{S}_y\|_F^{\frac{1}{2}}), \quad (29)$$

where the projection is defined as

$$\mathcal{P}_C(\mathbf{U}) = \arg \min_{\tilde{\mathbf{U}} \in \mathcal{C}} \left\| \mathbf{U} - \tilde{\mathbf{U}} \right\|_F, \quad (30)$$

that is,

$$[\mathcal{P}_C(\mathbf{U})]_{i,:} = \begin{cases} \mathbf{U}_{i,:}, & \text{if } \|\mathbf{U}_{i,:}\|_2 \leq 2\sqrt{\frac{\mu r \sigma}{n}}, \\ \frac{\mathbf{U}_{i,:}}{\|\mathbf{U}_{i,:}\|_2} 2\sqrt{\frac{\mu r \sigma}{n}}, & \text{otherwise,} \end{cases} \quad (31)$$

and $\mathbf{U}_{i,:}$ is the i -th row of $\mathbf{U}$.

Then we use gradient-based methods to update the iteration variable. Before providing the gradients, we review some important facts from [26].

Fact 1 (Proposition 3.4, [26]). *For a scalar function $h(\mathbf{S}, \bar{\mathbf{S}}) : \mathbb{C}^{m \times n} \times \mathbb{C}^{m \times n} \rightarrow \mathbb{C}$, if the matrix differential of the function is*

$$dh(\mathbf{S}, \bar{\mathbf{S}}) = \langle \mathbf{A}, d\mathbf{S} \rangle + \langle \mathbf{B}, d\bar{\mathbf{S}} \rangle, \quad (32)$$

then its complex gradient matrices are the conjugates of the matrices $\mathbf{A}$ and $\mathbf{B}$, i.e.,

$$\nabla_{\mathbf{S}} h(\mathbf{S}, \bar{\mathbf{S}}) = \bar{\mathbf{A}}, \text{ and } \nabla_{\bar{\mathbf{S}}} h(\mathbf{S}, \bar{\mathbf{S}}) = \bar{\mathbf{B}}. \quad (33)$$

Fact 2 (Theorem 4.4, [26]). *For a real function $h(\mathbf{S}, \bar{\mathbf{S}}) : \mathbb{C}^{m \times n} \times \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$, its fastest growing direction is given by $\nabla_{\bar{\mathbf{S}}} h(\mathbf{S}, \bar{\mathbf{S}})$.*

Based on Fact 2, the gradients of $\hat{F}_p(\mathbf{U})$, $\hat{f}_p(\mathbf{U})$ and $g(\mathbf{U})$ under Wirtinger calculus are

$$\nabla F_p(\mathbf{U}) \triangleq \nabla \hat{f}_p(\mathbf{U}) + \gamma \nabla g(\mathbf{U}), \quad (34)$$

$$\nabla \hat{f}_p(\mathbf{U}) \triangleq \nabla_{\bar{\mathbf{U}}} \hat{f}_p(\mathbf{U}), \quad (35)$$

$$\nabla g(\mathbf{U}) \triangleq \nabla_{\bar{\mathbf{U}}} g(\mathbf{U}). \quad (36)$$

It is easy to get the gradient of $g(\mathbf{U})$, which is

$$\nabla g(\mathbf{U}) = \nabla_{\bar{\mathbf{U}}} g(\mathbf{U}) = (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T) \bar{\mathbf{U}}. \quad (37)$$

Next, we provide the gradients of $\hat{f}_p(\mathbf{U})$ under Wirtinger calculus for $p = 1, 2$.

Lemma 1. *Define $\mathbf{w} = \text{Re}[\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)]$. Then*

(a) *For $p = 1$, the gradient of $\hat{f}_p(\mathbf{U})$ is*

$$\nabla \hat{f}_1(\mathbf{U}) = \mathcal{G} \mathcal{D}^{-1} (\phi'_b(\mathbf{w}) \odot \mathbf{y}) \bar{\mathbf{U}}. \quad (38)$$

(b) *For $p = 2$, the gradient of $\hat{f}_p(\mathbf{U})$ is*

$$\nabla \hat{f}_2(\mathbf{U}) = \mathcal{G} \mathcal{D}^{-1} (\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y}) \bar{\mathbf{U}}. \quad (39)$$

Proof. The proof is provided in Appendix E. $\square$

According to Lemma 1 and Eq. (37), we obtain

$$\nabla \hat{F}_1(\mathbf{U}) = \mathcal{G} \mathcal{D}^{-1} (\phi'_b(\mathbf{w}) \odot \mathbf{y}) \bar{\mathbf{U}} + \gamma (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T) \bar{\mathbf{U}}, \quad (40)$$

$$\nabla \hat{F}_2(\mathbf{U}) = \mathcal{G} \mathcal{D}^{-1} (\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y}) \bar{\mathbf{U}} + \gamma (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T) \bar{\mathbf{U}}. \quad (41)$$

Now, we are ready to describe our algorithm flow. The algorithm is provided in Algorithm 1. The maximum number of rounds is K . In each round k , the update rule is

$$\mathbf{U}^{k+1} = \mathcal{P}_C(\mathbf{U}^k - \eta \nabla \hat{F}_p(\mathbf{U}^k)), \quad (42)$$

where $\eta > 0$ is the step size. In the last round, we map the lifting matrix back to the signal

$$\mathbf{z}^K = \mathcal{D}^{-1} \mathcal{G}^* \left(\frac{\mathbf{U}^K (\mathbf{U}^K)^T}{\|\mathbf{U}^K (\mathbf{U}^K)^T\|_F} \right). \quad (43)$$

Algorithm 1 OSCAR- ℓ_p : One-Bit Spectral Compressed Sensing via Hankel Matrix Factorization

Input: $p, \gamma, \eta, \mathbf{y}$
Initialization: $\mathcal{T}_r(\mathcal{H}(\mathbf{y})) = \mathbf{U}_y \mathbf{S}_y \mathbf{U}_y^H, \mathbf{U}^0 = \mathcal{P}_{\mathcal{C}}(\mathbf{U}_y \mathbf{S}_y^{\frac{1}{2}} / \|\mathbf{S}_y\|_{\text{F}}^{\frac{1}{2}})$
for $k = 0, 1, \dots, K-1$ **do**
 $\mathbf{U}^{k+1} = \mathcal{P}_{\mathcal{C}}(\mathbf{U}^k - \eta \nabla \hat{F}_p(\mathbf{U}^k))$
end for
Output: $\mathbf{U}^K, \mathbf{z}^K = \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}^K (\mathbf{U}^K)^T / \|\mathbf{U}^K (\mathbf{U}^K)^T\|_{\text{F}})$

Empirical results and previous works [24], [27] demonstrate that normalizing the signal at the last round is enough for one-bit problems.

Based on Algorithm 1, we apply the momentum acceleration strategy to design the accelerated version in Algorithm 2

$$\mathbf{U}^{k+1} = \mathcal{P}_{\mathcal{C}'}(\tilde{\mathbf{U}}^k - \eta \nabla \hat{F}_p(\tilde{\mathbf{U}}^k)), \quad (44)$$

$$\tilde{\mathbf{U}}^{k+1} = \mathbf{U}^{k+1} + \beta(\mathbf{U}^{k+1} - \mathbf{U}^k), \quad (45)$$

where $\tilde{\mathbf{U}} \in \mathbb{C}^{s \times r}$ denotes the extrapolated matrix, $\beta > 0$ is an extrapolation parameter and the set $\mathcal{C}'$ is defined as

$$\mathcal{C}' = \left\{ \mathbf{U} \in \mathbb{C}^{s \times r} \mid \|\mathbf{U}\|_{2,\infty} \leq \frac{2}{1+2\beta} \sqrt{\frac{\mu r \sigma}{n}} \right\}. \quad (46)$$

Here we introduce a smaller set $\mathcal{C}'$ to ensure that the iterates $\mathbf{U}^k, \tilde{\mathbf{U}}^k$ fall into the incoherence set $\mathcal{C}$.

Besides, we modify the traditional accelerated method to ensure a sufficient decrease in the objective function, thereby achieving more stable convergence behavior [28], [29]. As shown in Algorithm 2, the update rule enforces a monotonic decrease by accepting the new iterate only when the objective function value is reduced.

Fast computation. We present the fast computation of the proposed algorithms. According to [18], [3], two kinds of terms can be computed quickly:

- 1) For any intermediate vector $\mathbf{q} \in \mathbb{C}^n$, the term $\mathcal{G}(\mathbf{q})\bar{\mathbf{U}}$ can be computed by fast Hankel matrix-vector multiplication, which is realized by r fast convolutions with $O(nr \log(n))$ flops via FFT.
- 2) The term $\mathcal{G}^*(\mathbf{U}\mathbf{U}^T)$ can be computed quickly by r fast convolution with $O(nr \log(n))$ flops via FFT.

The computational complexity for (37) is also $O(nr \log(n))$ [18], [3]. Additionally, the projection operation can be integrated into the computation involved in fast convolution, thus eliminating the need to separately account for the computational complexity of projecting onto $\mathcal{C}$. As a result, the computational complexity of each iteration of our algorithms is $O(nr \log(n))$.

V. CONVERGENCE ANALYSIS

This section gives the convergence analysis for Algorithm 1 and Algorithm 2. Before giving the theoretical analysis, we first introduce some useful facts and lemmas. Subsequently, we will provide the Hessian matrices and prove the smoothness of the objective functions. Finally, we analyze the convergence properties of the two algorithms.

Algorithm 2 Acc-OSCAR- ℓ_p : Accelerated One-Bit Spectral Compressed Sensing via Hankel Matrix Factorization

Input: $p, \gamma, \eta, \beta, \mathbf{y}$
Initialization: $\mathcal{T}_r(\mathcal{H}(\mathbf{y})) = \mathbf{U}_y \mathbf{S}_y \mathbf{U}_y^H, \mathbf{U}^0 = \tilde{\mathbf{U}}^0 = \mathcal{P}_{\mathcal{C}'}(\mathbf{U}_y \mathbf{S}_y^{\frac{1}{2}} / \|\mathbf{S}_y\|_{\text{F}}^{\frac{1}{2}})$
for $k = 0, 1, \dots, K-1$ **do**
 $\mathbf{U}^{k+1} = \mathcal{P}_{\mathcal{C}'}(\tilde{\mathbf{U}}^k - \eta \nabla \hat{F}_p(\tilde{\mathbf{U}}^k))$
 $\mathbf{V}^{k+1} = \mathbf{U}^{k+1} + \beta(\mathbf{U}^{k+1} - \mathbf{U}^k)$
if $\hat{F}_p(\mathbf{U}^{k+1}) \leq \hat{F}_p(\mathbf{V}^{k+1})$ **then**
 $\tilde{\mathbf{U}}^{k+1} = \mathbf{U}^{k+1}$
else if $\hat{F}_p(\mathbf{V}^{k+1}) < \hat{F}_p(\mathbf{U}^{k+1})$ **then**
 $\tilde{\mathbf{U}}^{k+1} = \mathbf{V}^{k+1}$
end for
Output: $\mathbf{U}^K, \mathbf{z}^K = \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}^K (\mathbf{U}^K)^T / \|\mathbf{U}^K (\mathbf{U}^K)^T\|_{\text{F}})$

Define the orthonormal bases for $s \times s$ Hankel matrix

$$\mathbf{H}_a = \frac{1}{\sqrt{w_a}} \mathcal{H}(\mathbf{e}_a), a = 1, \dots, n. \quad (47)$$

For any Hankel matrix $\mathcal{H}(\mathbf{z})$, we have

$$\mathcal{H}(\mathbf{z}) = \sum_{a=1}^n \langle \mathbf{H}_a, \mathcal{H}(\mathbf{z}) \rangle \mathbf{H}_a. \quad (48)$$

In addition, $\mathcal{G}\mathcal{G}^*(\mathbf{W})$ can be reformulated as

$$\begin{aligned} \mathcal{G}\mathcal{G}^*(\mathbf{W}) &= \sum_{a=1}^n \langle \mathbf{H}_a, \mathcal{G}\mathcal{G}^*(\mathbf{W}) \rangle \mathbf{H}_a \\ &= \sum_{a=1}^n \langle \mathcal{G}\mathcal{G}^*(\mathbf{H}_a), \mathbf{W} \rangle \mathbf{H}_a \\ &= \sum_{a=1}^n \langle \mathbf{H}_a, \mathbf{W} \rangle \mathbf{H}_a. \end{aligned} \quad (49)$$

Next, we review one important fact about the relationship between the second-order matrix differential and the Hessian matrix, which is reformulated for our proof.

Fact 3 (Theorem 3.4, [26]). *For a differentiable real function $h(\mathbf{S}, \bar{\mathbf{S}}) : \mathbb{C}^{m \times n} \times \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$, if its second-order matrix differential form is*

$$\begin{aligned} d^2 h(\mathbf{S}, \bar{\mathbf{S}}) &= (\text{dvec}(\mathbf{S}))^T \mathbf{C} \text{dvec}(\mathbf{S}) \\ &\quad + (\text{dvec}(\bar{\mathbf{S}}))^T \mathbf{D} \text{dvec}(\bar{\mathbf{S}}) \\ &\quad + (\text{dvec}(\mathbf{S}))^T \bar{\mathbf{D}} \text{dvec}(\bar{\mathbf{S}}) \\ &\quad + (\text{dvec}(\bar{\mathbf{S}}))^T \bar{\mathbf{C}} \text{dvec}(\bar{\mathbf{S}}), \end{aligned} \quad (50)$$

then the full Hessian matrix of $h(\mathbf{S}, \bar{\mathbf{S}})$ is

$$\nabla^2 h(\mathbf{S}, \bar{\mathbf{S}}) = \begin{bmatrix} \mathbf{H}_{\bar{\mathbf{S}}, \mathbf{S}} & \mathbf{H}_{\bar{\mathbf{S}}, \bar{\mathbf{S}}} \\ \mathbf{H}_{\mathbf{S}, \mathbf{S}} & \mathbf{H}_{\mathbf{S}, \bar{\mathbf{S}}} \end{bmatrix}, \quad (51)$$

where

$$\mathbf{H}_{\mathbf{S}, \mathbf{S}} = \frac{1}{2}(\mathbf{C} + \mathbf{C}^T), \quad \mathbf{H}_{\mathbf{S}, \bar{\mathbf{S}}} = \frac{1}{2}(\mathbf{D} + \mathbf{D}^H), \quad (52)$$

$$\mathbf{H}_{\bar{\mathbf{S}}, \bar{\mathbf{S}}} = \overline{\mathbf{H}_{\mathbf{S}, \mathbf{S}}}, \quad \mathbf{H}_{\bar{\mathbf{S}}, \mathbf{S}} = \overline{\mathbf{H}_{\mathbf{S}, \bar{\mathbf{S}}}}. \quad (53)$$

Then we provide the Hessian matrices for $\hat{F}_1(\mathbf{U})$ and $\hat{F}_2(\mathbf{U})$, respectively.

Lemma 2. Define $\mathbf{w} = \text{Re}(\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T))$. Then we have:

(a) For $p = 1$, the Hessian matrix of $\hat{F}_p(\mathbf{U})$ is

$$\nabla^2 \hat{F}_1(\mathbf{U}) = \begin{bmatrix} \mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{F}_1} & \mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{F}_1} \\ \mathbf{H}_{\mathbf{U}, \mathbf{U}}^{\hat{F}_1} & \mathbf{H}_{\mathbf{U}, \bar{\mathbf{U}}}^{\hat{F}_1} \end{bmatrix} \quad (54)$$

with

$$\begin{aligned} \mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{F}_1} &= (\mathbf{U}^T \otimes \mathbf{I}_s) \mathbf{\Gamma}_f \text{diag}(\mathcal{G} \mathcal{D}^{-1}(\phi_b''(\mathbf{w}))) (\bar{\mathbf{U}} \otimes \mathbf{I}_s) \\ &\quad + \gamma (\mathbf{U}^T \otimes \mathbf{I}_s) \mathbf{\Gamma}_g (\bar{\mathbf{U}} \otimes \mathbf{I}_s), \end{aligned} \quad (55)$$

$$\begin{aligned} \mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{F}_1} &= \mathbf{I}_r \otimes \mathcal{G} \mathcal{D}^{-1}(\phi_b'(\mathbf{w}) \odot \mathbf{y}) + \frac{1}{2} (\mathbf{U}^H \otimes \mathbf{I}_s) \mathbf{\Gamma}_f \\ &\quad \cdot \text{diag}(\mathcal{G} \mathcal{D}^{-1}(\phi_b''(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y})) (\bar{\mathbf{U}} \otimes \mathbf{I}_s) \\ &\quad + \gamma \mathbf{I}_r \otimes (\mathcal{I} - \mathcal{G} \mathcal{G}^*)(\mathbf{U}\mathbf{U}^T), \end{aligned} \quad (56)$$

$$\mathbf{H}_{\mathbf{U}, \bar{\mathbf{U}}}^{\hat{F}_1} = \overline{\mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{F}_1}}, \quad \mathbf{H}_{\mathbf{U}, \mathbf{U}}^{\hat{F}_1} = \overline{\mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{F}_1}}, \quad (57)$$

where

$$\mathbf{\Gamma}_g = (\mathbf{I}_{s^2} + \mathbf{K}_{s^2}) \left(\mathbf{I}_{s^2} - \sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T \right), \quad (58)$$

$$\mathbf{\Gamma}_f = (\mathbf{I}_{s^2} + \mathbf{K}_{s^2}) \left(\sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T \right), \quad (59)$$

and $\mathbf{K}_{s^2} \in \mathbb{R}^{s^2 \times s^2}$ denotes the perfect shuffle matrix that can convert column-major vectorization to row-major vectorization [30].

(b) For $p = 2$, the Hessian matrix of $\hat{F}_p(\mathbf{U})$ is

$$\nabla^2 \hat{F}_2(\mathbf{U}) = \begin{bmatrix} \mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{F}_2} & \mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{F}_2} \\ \mathbf{H}_{\mathbf{U}, \mathbf{U}}^{\hat{F}_2} & \mathbf{H}_{\mathbf{U}, \bar{\mathbf{U}}}^{\hat{F}_2} \end{bmatrix} \quad (60)$$

with

$$\begin{aligned} \mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{F}_2} &= (\mathbf{U}^T \otimes \mathbf{I}_s) \mathbf{\Gamma}_f \text{diag}(\mathcal{G} \mathcal{D}^{-1}(\Lambda_\phi(\mathbf{w}))) (\bar{\mathbf{U}} \otimes \mathbf{I}_s) \\ &\quad + \gamma (\mathbf{U}^T \otimes \mathbf{I}_s) \mathbf{\Gamma}_g (\bar{\mathbf{U}} \otimes \mathbf{I}_s), \end{aligned} \quad (61)$$

$$\begin{aligned} \mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{F}_2} &= \mathbf{I}_r \otimes \mathcal{G} \mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi_b'(\mathbf{w}) \odot \mathbf{y}) + \frac{1}{2} (\mathbf{U}^H \otimes \mathbf{I}_s) \\ &\quad \cdot \mathbf{\Gamma}_f \text{diag}(\mathcal{G} \mathcal{D}^{-1}(\Lambda_\phi(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y})) (\bar{\mathbf{U}} \otimes \mathbf{I}_s) \\ &\quad + \gamma \mathbf{I}_r \otimes (\mathcal{I} - \mathcal{G} \mathcal{G}^*)(\mathbf{U}\mathbf{U}^T), \end{aligned} \quad (62)$$

$$\mathbf{H}_{\mathbf{U}, \bar{\mathbf{U}}}^{\hat{F}_2} = \overline{\mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{F}_2}}, \quad \mathbf{H}_{\mathbf{U}, \mathbf{U}}^{\hat{F}_2} = \overline{\mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{F}_2}}, \quad (63)$$

where

$$\Lambda_\phi(\mathbf{w}) = \phi_b'(\mathbf{w}) \odot \phi_b'(\mathbf{w}) + \phi_b(\mathbf{w}) \odot \phi_b''(\mathbf{w}). \quad (64)$$

Proof. The proof is provided in Appendix J of the Supplementary Material. $\square$

Next, we present that the objective function is smooth by proving that the Hessian matrices are upper bounded.

Lemma 3. For any $\mathbf{U} \in \mathcal{C}$, where $\mathcal{C}$ is the set defined in (16), the spectral norm of $\nabla^2 \hat{F}_p(\mathbf{U})$ is upper-bounded, that is,

$$\|\nabla^2 \hat{F}_p(\mathbf{U})\| \leq L_p, \quad (65)$$

where L_p is defined as

$$L_p = \begin{cases} 2\sqrt{2n} + 8\mu r \sigma(\frac{2}{\sqrt{b}} + 3\gamma), & p = 1, \\ 8(\sqrt{\mu^2 r^2 \sigma^2 / n + n b} + \mu r \sigma / \sqrt{n}) + \mu r \sigma(9 + 3\gamma), & p = 2. \end{cases}$$

Proof. The proof is provided in Appendix F. $\square$

Finally, we theoretically establish the convergence of the algorithms toward an approximate first-order stationary point. Based on the update rule in Eqs. (42) and (44), the first-order optimality condition [31], [32] means that there exists $\mathbf{U} \in \mathbb{C}^{n \times r}$ such that

$$-\nabla \hat{F}_p(\mathbf{U}) \in N_{\mathcal{C}}(\mathbf{U}), \quad (66)$$

and the approximate first-order optimality condition means that there exists $\mathbf{U} \in \mathbb{C}^{n \times r}$ such that

$$-\nabla \hat{F}_p(\mathbf{U}) \in N_{\mathcal{C}}(\mathbf{U}) + \mathcal{B}_2(0, \varepsilon), \quad (67)$$

where $N_{\mathcal{C}}(\mathbf{U})$ denotes the normal cone of $\mathcal{C}$ at $\mathbf{U}$ and $\mathcal{B}_2(\mathbf{z}, r) \triangleq \{\mathbf{q} \in \mathbb{C}^d : \|\mathbf{q} - \mathbf{z}\|_2 \leq r\}$ denotes the Euclidean ball of radius r centered at $\mathbf{z}$.

Theorem 1. For the iterates in Algorithm 1, the limit of the loss function satisfies

$$\lim_{k \rightarrow \infty} \hat{F}_p(\mathbf{U}^k) = \hat{F}_p^*, \quad (68)$$

and the sequence of iterates satisfies

$$\min_{k \in [K]} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F \leq \sqrt{\frac{2\hat{F}_p(\mathbf{U}^0)}{L_p K}}, \quad (69)$$

as long as $0 < \eta \leq \frac{1}{L_p}$, where $\hat{F}_p^*$ denotes some positive constant.

Proof. The proof is presented in Appendix A. $\square$

Theorem 2. Assume that K is large enough to satisfy

$$K \geq \frac{8L_p \hat{F}_p(\mathbf{U}^0)}{\varepsilon^2}, \quad (70)$$

and $\eta = \frac{1}{L_p}$, then there exists $k' \in [K]$ in Algorithm 1 such that it satisfies the approximate first-order optimality condition

$$-\nabla \hat{F}_p(\mathbf{U}^{k'+1}) \in N_{\mathcal{C}}(\mathbf{U}^{k'+1}) + \mathcal{B}_2(0, \varepsilon). \quad (71)$$

Proof. The proof is shown in Appendix B. $\square$

Remark 2. Theorem 1 establishes that the value of the function in our algorithm converges to a certain limit, with the minimum of the difference $\mathbf{U}^{k+1} - \mathbf{U}^k$ having an upper bound. Furthermore, Theorem 2 determines the iteration complexity required to satisfy the approximate first-order optimality condition, which is inversely proportional to the square of the error ε . Moreover, the iteration complexity is directly proportional to the initial function value $\hat{F}_p(\mathbf{U}^0)$. Therefore, effective initialization contributes to faster convergence to the approximate first-order optimality condition.

Theorem 3. Assuming $\eta = \frac{1}{L_p}$, the sequence $\{\mathbf{U}^k\}$ generated by Algorithm 2 satisfies

$$\lim_{k \rightarrow \infty} \hat{F}_p(\mathbf{U}^k) = \hat{F}_p^\circ. \quad (72)$$

Besides, the set of limit points Ω of $\{U^k\}$ forms a compact set. For $U' \in \Omega$, it satisfies the first-order optimality condition that

$$-\nabla \hat{F}_p(U') \in N_C(U').$$

Proof. See Appendix C. $\square$

Definition 1 (Distance). The distance of a point $x \in \mathbb{C}^d$ to a closed set $\Theta \subseteq \mathbb{C}^d$ is defined as

$$\text{dist}_\Theta(x) = \min_{y \in \Theta} \|y - x\|_2.$$

We introduce the Kurdyka-Lojasiewicz (KL) property [33][28], which is important to derive the convergence rate of the accelerated algorithm (Algorithm 2).

Assumption 1. The function h is said to satisfy the uniformized Kurdyka-Lojasiewicz (KL) property [33][28] if

$$\varphi'(h(x) - h^*) \text{dist}_{\partial h(x)}(\mathbf{0}) \geq \frac{1}{c_{\text{KL}}},$$

where c_{KL} is the KL constant, $\varphi(t) = 2t^{\frac{1}{2}}$, and h^* is some limit value of the function h .

Finally, we introduce an auxiliary function that plays an important role in establishing the convergence rate property of Algorithm 2.

$$G(U) = \hat{F}_p(U) + I_{C'}(U),$$

where $U \in \mathbb{C}^{n \times r}$, $I_{C'}$ is an indicator function such that $I_{C'}(U) = \infty$ for $U \notin C'$ and $I_{C'}(U) = 0$ for $U \in C'$.

Theorem 4. Suppose $G(U^k)$ satisfies the KL property in Assumption 1 for $k \in [K]$:

$$\varphi'(G(U^k) - \hat{F}_p^\diamond) \text{dist}_{\partial G(U^k)}(\mathbf{0}) \geq \frac{1}{c_{\text{KL}}}, \quad (73)$$

where $\hat{F}_p^\diamond$ is the limit value defined in (72). For $\eta = \frac{1}{L_p}$, the sequence of iterations in Algorithm 2 satisfies

$$\hat{F}_p(U^k) - \hat{F}_p^\diamond \leq \left(1 - \frac{1}{1 + C_p}\right)^k (\hat{F}_p(U^0) - \hat{F}_p^\diamond),$$

where $C_p = 8c_{\text{KL}}^2 L_p$.

Proof. See Appendix D. $\square$

Remark 3. Theorem 3 establishes that the value of the function in Algorithm 2 converges to a certain limit, and the set of limit points satisfies the first-order optimality condition. Theorem 4 gives the linear convergence rate of Algorithm 2 under the KL assumption. It is generally difficult to analyze the KL property of an arbitrary function, since it is often unclear whether the function is semi-algebraic [29], [34]. Moreover, the analysis of the KL property relies on the explicit expression of $\text{dist}_{\partial h(x)}(\mathbf{0})$ in Assumption 1, which is challenging to derive in practice. Therefore, we instead verify the stable KL property of Algorithm 2 empirically through simulations, as reported in Appendix G of the Supplementary Material.

Remark 4 (Scope of convergence guarantees and local minima). Theorems 2 and 3 establish convergence to a first-order stationary point of the smooth objective (25); they do not theoretically guarantee recovery of the true frequency

locations. Frequency recovery is demonstrated empirically in Figs. 6-8 but remains an open theoretical problem.

Regarding local minima, the problem shares the low-rank factorization structure studied by Ge et al. [35], who proved that all local minima are global for that class; this provides indirect evidence that the stationary points reached by OSCAR and Acc-OSCAR are of high quality. Empirically, Acc-OSCAR matches the RMSE of the convex benchmark 1-bit ANM [22] in Figs. 6-8, which is globally optimal but far more expensive. Performance degrades only in three identifiable regimes: low SNR, insufficient frequency separation, or poor initialization (see Figs. 9 and 10 for more information).

VI. SIMULATIONS

This section provides the numerical simulations on one-bit DoA estimation with one snapshot for the proposed OSCAR algorithms and then compares the two best of them with previous advanced algorithms, including CBIHT- ℓ_1 [4], CBIHT- ℓ_2 [4], iCBIHT [5], 1bit-ANM [22], and DeepFPC [11].

DoA estimation studies the problem of estimating the angle of targets by the received signals of multiple sensors. Considering the DoA estimation of a uniform linear array of n sensors

$$x_t^* = \sum_{a=1}^r d_a \exp\left(-i2\pi t \sin(\theta_a) \frac{d}{\lambda}\right), t = 1, 2, \dots, n, \quad (74)$$

where $\theta_a \in [-90^\circ, 90^\circ)$ denotes the direction of arrivals and $\theta = [\theta_1, \theta_2, \dots, \theta_r]$, d denotes the spacing between two adjacent sensors, and λ denotes the wavelength of the signals. Notice that spectral compressed sensing becomes DoA estimation by setting $f_a = \frac{(\sin(\theta_a)+1)d}{\lambda}$, $\tau_a = \frac{d}{\lambda}$, and $d = \frac{\lambda}{2}$ in Eq. (1). The performance is assessed under Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{r} \sum_{a=1}^r \|\hat{\theta}_a - \theta_a\|_2^2}, \quad (75)$$

where $\hat{\theta}_a$ denotes the estimation of θ_a .

A. Comparisons of the proposed OSCAR algorithms

In this section, we first compare the performance of OSCAR- ℓ_1 , OSCAR- ℓ_2 , Acc-OSCAR- ℓ_1 , Acc-OSCAR- ℓ_2 , Acc-COSCAR- ℓ_1 , and Acc-COSCAR- ℓ_2 in Figs. 3 and 4. Here, Acc-COSCAR- ℓ_1 and Acc-COSCAR- ℓ_2 correspond to the accelerated algorithms Acc-OSCAR- ℓ_1 , Acc-OSCAR- ℓ_2 by replacing $\text{Re}(\bar{y} \odot z)$ with

$$\begin{pmatrix} \text{Re}(\mathbf{y}) \odot \text{Re}(\mathbf{z}) \\ \text{Im}(\mathbf{y}) \odot \text{Im}(\mathbf{z}) \end{pmatrix}. \quad (76)$$

For brevity, we omit the related details. For the parameters in the projection set $\mathcal{C}$ of our algorithms, we set μ as $\mu = 1$ and σ is determined as $\sigma = \sigma_1(\mathbf{U}_y \mathbf{S}_y \mathbf{U}_y^H / \|\mathbf{S}_y\|_F)$.

The dimension of $\mathbf{x}^*$ is set to be $n = 63$, the number of sinusoids is $r = 2$, $\theta = [20.3^\circ, 22.5^\circ]$, the tradeoff parameter is $\gamma = 100$, the step size is $\eta = 0.001$, the smoothness parameter is $b = 0.001$, the momentum parameter is $\beta = 0.7$, and the maximum number of iterations is $K = 100$. One hundred Monte Carlo trials were conducted to get the RMSE.

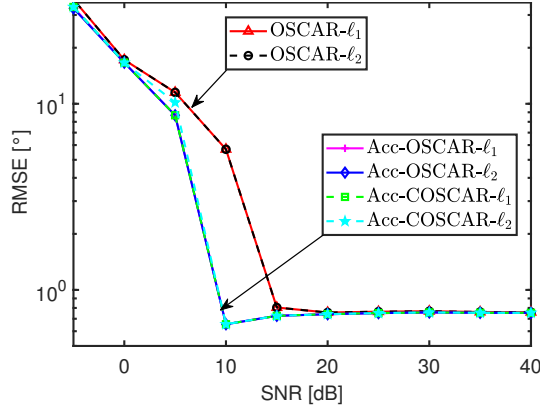


Fig. 3. RMSE vs SNR for all OSCARS.

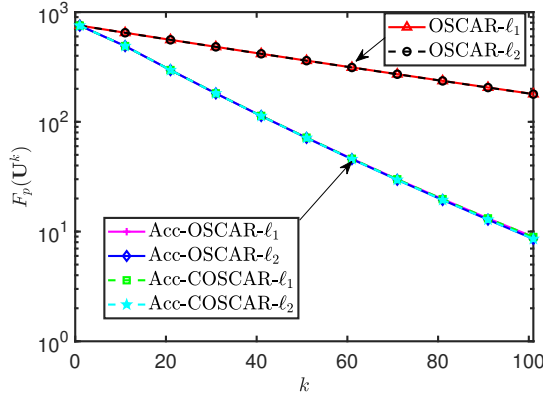


Fig. 4. Objective function $F_p(\mathbf{U}^k)$ vs k for all OSCARS.

Fig. 3 shows the RMSE as a function of SNR and Fig. 4 presents the objective function $F_p(\mathbf{U}^k)$ as a function of k for all OSCAR algorithms for SNR = 10 dB. The results indicate that the performance of all accelerated algorithms is nearly identical, and they exceed the performance of non-accelerated algorithms. This observation corroborates the statement in Remark 1 and confirms the efficacy of the acceleration technique.

It is worth noting that an error floor appears at high SNR in Fig. 3. The error floor observed at high SNR is a direct consequence of one-bit quantization under a single-snapshot setting: the amplitude information is irreversibly lost, so the estimation error cannot vanish even as the noise level becomes negligible. To further investigate whether this floor reflects a fundamental limitation or merely a finite-dimensional effect, Fig. 5 reports the RMSE of Acc-OSCAR for several odd values of n , specifically $n = 63$, $n = 95$, and $n = 127$. It can be observed that the error floor decreases monotonically as n increases, indicating that it is not a universal constant but rather a finite-dimensional artifact. From a structural perspective, increasing n enhances the redundancy of the lifted Hankel matrix, thereby strengthening the low-rank constraint and improving the stability of gridless frequency localization.

B. DOA estimation with one snapshot

Since Acc-OSCAR- ℓ_1 and Acc-OSCAR- ℓ_2 outperform other OSCAR algorithms, we compare our Acc-OSCAR- ℓ_1

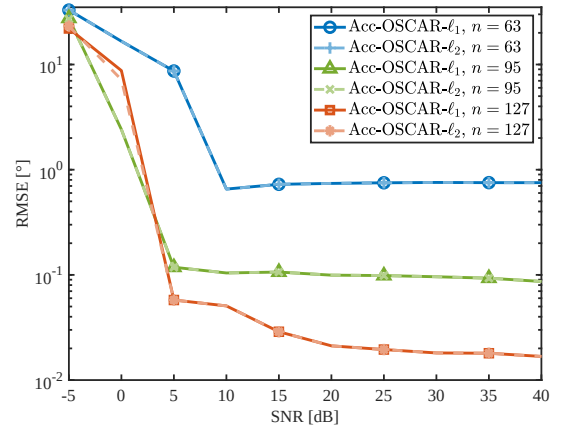
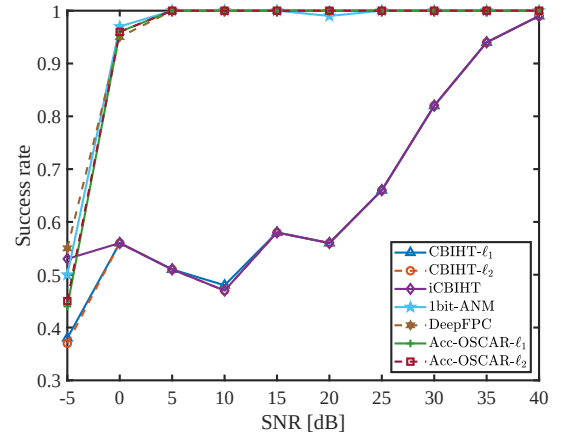
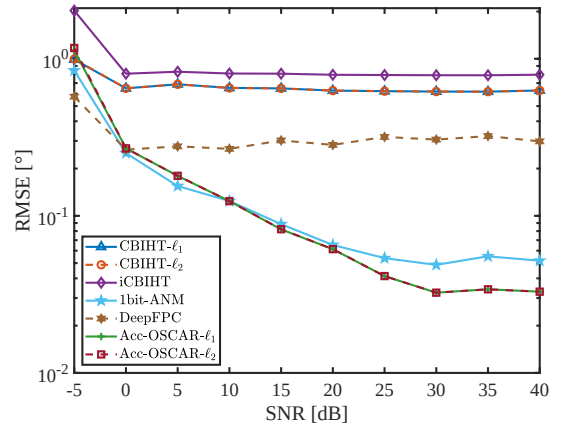


Fig. 5. RMSE vs SNR for Acc-OSCAR- ℓ_1 and Acc-OSCAR- ℓ_2 with different signal dimensions n . It is observed that the high-SNR RMSE floor becomes lower as n increases.



(a)

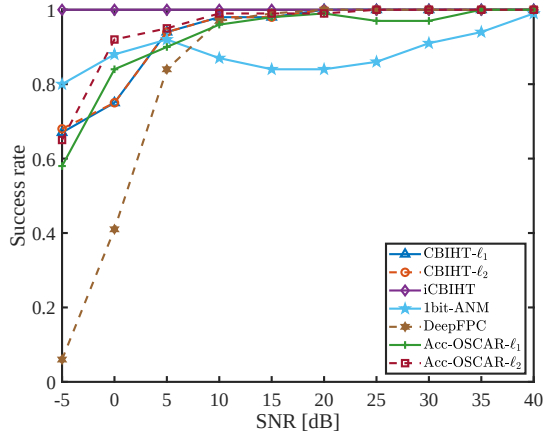


(b)

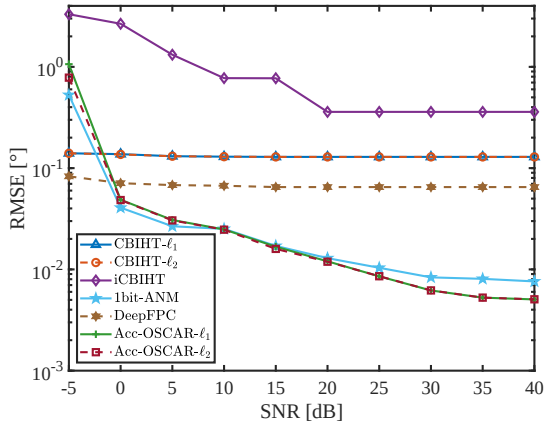
Fig. 6. Comparisons for different algorithms with $n = 63$, $r = 2$, and $\theta = [-20.3^\circ, 24.5^\circ]$. (a) Success Rate vs SNR. (b) RMSE vs SNR.

and Acc-OSCAR- ℓ_2 with CBIHT, iCBIHT, 1bit-ANM, and DeepFPC in this subsection. In particular, DeepFPC adopts an 8-layer unfolded architecture, trained on 10,000 samples over 1,000 epochs. The parameters for other algorithms are set as recommended.

First of all, we compare the success rate and the RMSE for



(a)



(b)

Fig. 7. Comparisons for different algorithms with $n = 255$, $r = 3$ and $\theta = [-56.3^\circ, -15.4^\circ, 30.5^\circ]$. (a) Success Rate vs SNR. (b) RMSE vs SNR.

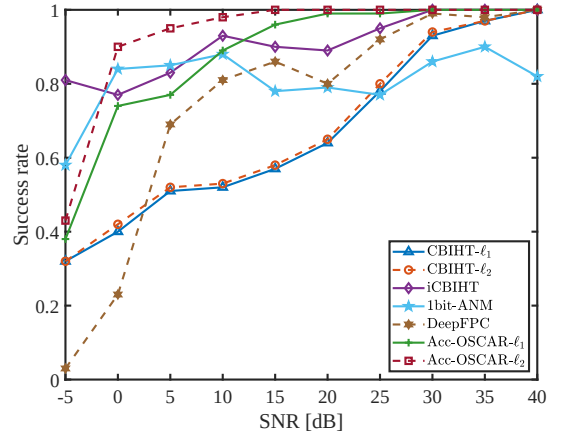
all successful trials of all algorithms with $n = 63$, $r = 2$, and $\theta = [-20.3^\circ, 24.5^\circ]$, and the maximum number of iterations is $K = 100$. A trial is called successful if its RMSE satisfies

$$\text{RMSE} < 3^\circ. \quad (77)$$

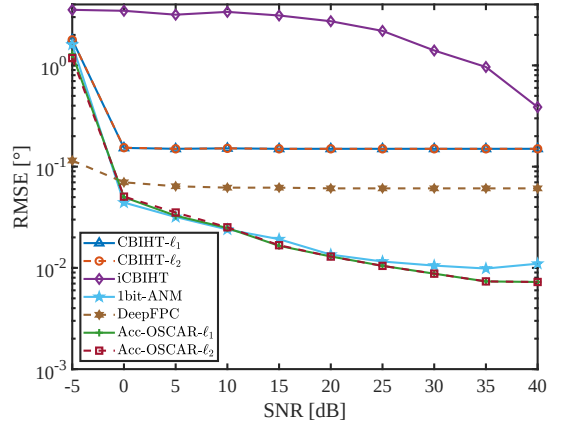
One hundred Monte Carlo trials were conducted to get the success rate and RMSE. The rationale behind introducing the success rate is that there is a possibility that the DoAs cannot be recovered in certain trials under a one-bit single-measurement vector and low SNR. We carefully choose a relatively small number 3° to facilitate easy comparison in the plot.

Fig. 6(a) shows that our algorithms Acc-OSCAR- ℓ_1 and Acc-OSCAR- ℓ_2 have similar success rates to 1-bit ANM and DeepFPC, which perform much better than CBIHT- ℓ_1 , CBIHT- ℓ_2 and iCBIHT. Fig. 6(b) shows that Acc-OSCAR- ℓ_1 and Acc-OSCAR- ℓ_2 have smaller RMSE than other algorithms, especially when the SNR is larger than 20 dB.

Then we conduct simulations for a large number of sensors, where $n = 255$, $r = 3$ and $\theta = [-56.3^\circ, -15.4^\circ, 30.5^\circ]$, and the maximum number of iterations is $K = 100$. As illustrated in Fig. 7, Acc-OSCAR- ℓ_2 demonstrates superior performance compared to other advanced algorithms, since Acc-OSCAR- ℓ_2 has a better success rate curve except for the iCBIHT



(a)



(b)

Fig. 8. Comparisons for different algorithms with $n = 255$, $r = 4$, and $\theta = [-56.3^\circ, -15.4^\circ, 10.5^\circ, 30.8^\circ]$. (a) Success Rate vs SNR. (b) RMSE vs SNR.

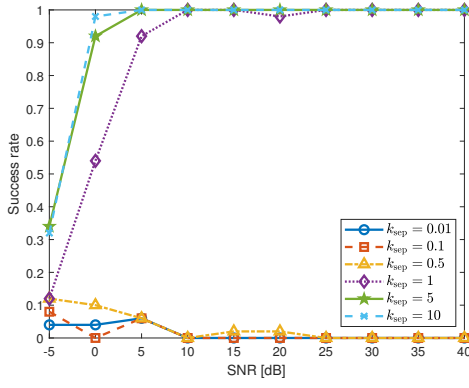
algorithm, and nearly achieves the best RMSE. The enhanced performance of Acc-OSCAR- ℓ_2 over Acc-OSCAR- ℓ_1 can be attributed to the nature of the noise, which is dense rather than sparse. Additionally, although iCBIHT has a 100% success rate, it suffers from poor RMSE since iCBIHT is designed for multiple snapshots, whereas this particular setting involves only a single snapshot.

To further evaluate the scalability of the proposed method with respect to the model order, we additionally consider the case of $n = 255$ and $r = 4$. The target directions are set to $\theta = [-56.3^\circ, -15.4^\circ, 10.5^\circ, 30.8^\circ]$. The corresponding success rate and RMSE results are shown in Fig. 8. It can be observed that the proposed method maintains competitive success rates and achieves lower RMSE compared to the baseline methods across most SNR levels. These results further demonstrate the robustness and effectiveness of the proposed algorithm in higher-order spectral estimation scenarios.

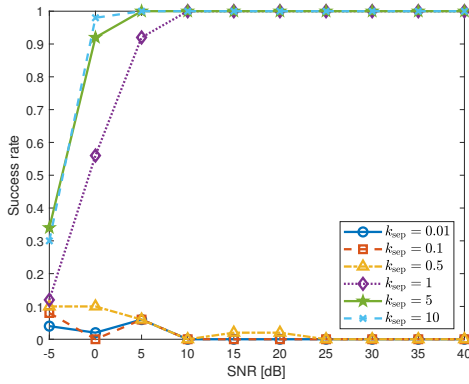
Table I reports the average runtime per trial for different algorithms, measured in milliseconds. It can be observed that the computational cost of optimization-based methods increases as the problem dimension grows. In particular, 1bit-ANM requires 500ms when $n = 63$ and its runtime increases dramatically to over 5×10^4 ms for $n = 255$, indicating limited scalability. In contrast, Acc-OSCAR and

TABLE I
AVERAGE RUNTIME PER TRIAL (MILLISECONDS).

Method	CBIHT- ℓ_1	CBIHT- ℓ_2	iCBIHT	1bit-ANM	Acc-OSCAR- ℓ_1	Acc-OSCAR- ℓ_2	DeepFPC
$n = 63, r = 2$	4.26	4.38	2.93	5.00×10^2	3.91	3.99	5.0×10^{-3}
$n = 255, r = 3$	22.03	22.16	7.52	5.10×10^4	21.46	21.64	5.2×10^{-3}



(a) Acc-OSCAR- ℓ_1



(b) Acc-OSCAR- ℓ_2

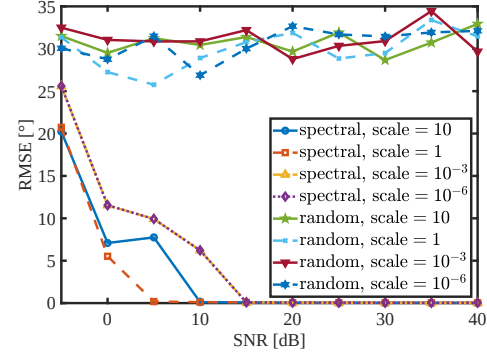
Fig. 9. Impact of different frequency separations on reconstruction performance with $n = 63, r = 2$. (a) Acc-OSCAR- ℓ_1 . (b) Acc-OSCAR- ℓ_2 . We set the minimum distance between the frequencies $\{f_a\}_{a=1}^r$ as $\frac{k_{\text{sep}}}{n}$, where the frequency separation parameter is chosen as $k_{\text{sep}} = 0.01, 0.1, 0.5, 1, 5, 10$.

CBIHT exhibit comparable runtime behavior, remaining on the order of a few milliseconds for smaller problem sizes and tens of milliseconds for larger dimensions. By comparison, DeepFPC achieves significantly lower inference runtime on the order of 10^{-3} ms. This result highlights the efficiency advantage of forward-pass inference, while also reflecting the trade-off that learning-based methods rely on offline training and are tailored to fixed problem dimensions.

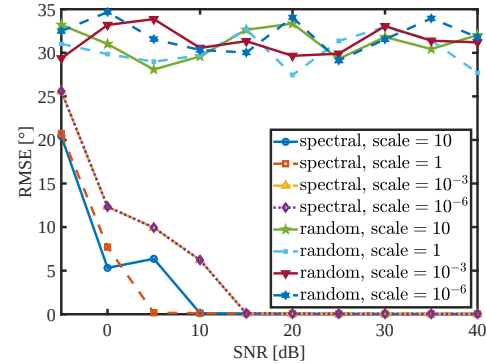
C. Robustness to Frequency Separation and Initialization

In addition to the SNR experiments in Figs. 6-8, we further investigate the robustness of the proposed Acc-OSCAR algorithms under more challenging scenarios, including reduced minimum frequency separation and different initialization strategies.

Frequency separation. The minimum wrap-around frequency separation k_{sep}/n controls the conditioning of the



(a) Acc-OSCAR- ℓ_1



(b) Acc-OSCAR- ℓ_2

Fig. 10. Impact of different initialization strategies on reconstruction performance with $n = 63, r = 2$ and $\theta = [-20.3^\circ, 24.5^\circ]$. (a) Acc-OSCAR- ℓ_1 . (b) Acc-OSCAR- ℓ_2 . The RMSE is plotted versus SNR for spectral and random initializations with different scaling factors.

lifting Hankel matrix $\mathcal{H}(x^*)$. When two frequencies are too close, the corresponding columns of the Vandermonde factor become nearly collinear, making spectral separation inherently difficult. Fig. 9 shows the success rate of Acc-OSCAR- ℓ_1 and Acc-OSCAR- ℓ_2 versus SNR for separation parameters $k_{\text{sep}} \in \{0.01, 0.1, 0.5, 1, 5, 10\}$ ($n = 63, r = 2$). When k_{sep} is very small, e.g., $k_{\text{sep}} = 0.01, 0.1, 0.5$, the two algorithms fail even at high SNR, as the Hankel matrix becomes ill-conditioned. Once k_{sep} is relatively large, both algorithms achieve high success rates above a moderate SNR threshold and the performance stabilizes.

Initialization strategies. We further evaluate the influence of different initialization strategies. Fig. 10 compares spectral initialization and random initialization across different scaling factors for $n = 63, r = 2$ and $\theta = [-20.3^\circ, 24.5^\circ]$. Specifically, given the spectral initial point U^0 (or a random standard Gaussian counterpart), the actual starting iterate is taken as ζU^0 for $\zeta \in \{10^{-6}, 10^{-3}, 1, 10\}$, so as to assess sensitivity to the magnitude of the initialization. Spectral initialization consistently yields near-zero RMSE once the SNR exceeds a

moderate threshold and is robust to the choice of scaling factor. In contrast, random initialization leads to substantially higher RMSE across all SNR levels, indicating that the algorithm is trapped in a suboptimal local minimum. This gap confirms that spectral initialization is a critical component of the proposed framework: it exploits the subspace information encoded in the sign-consistent Hankel matrix $\mathcal{H}(\mathbf{y})$ to provide a structurally meaningful starting point, enabling the subsequent gradient iterations to converge to a high-quality solution.

Overall, these experiments show that adequate frequency separation and spectral initialization are both necessary for reliable recovery, and that the proposed algorithms perform well when these conditions are satisfied.

VII. CONCLUSIONS

This paper proposes a gridless one-bit single-measurement spectral compressed sensing algorithm based on low-rank Hankel matrix factorization. We establish a non-convex optimization model using the low-rank property of the Hankel matrix and one-sided ℓ_p norm. A gradient descent algorithm, OSCAR, is designed with the smoothing technique, and its accelerated version, Acc-OSCAR, is designed by using momentum techniques. Theoretical analysis and experiments validate its convergence in single-measurement scenarios, confirming the algorithm's effectiveness.

APPENDIX A

PROOF OF THEOREM 1

Note that $\mathbf{U}^{k+1}$ is the solution to the following optimization problem.

$$\mathbf{U}^{k+1} = \arg \min_{\mathbf{U} \in \mathcal{C}} \langle \nabla \hat{F}_p(\mathbf{U}^k), \mathbf{U} - \mathbf{U}^k \rangle + \frac{1}{2\eta} \|\mathbf{U} - \mathbf{U}^k\|_F^2 \quad (78)$$

$$= \mathcal{P}_{\mathcal{C}}(\mathbf{U}^k - \eta \nabla \hat{F}_p(\mathbf{U}^k)), \quad (79)$$

For all $\mathbf{U} \in \mathcal{C}$, we have

$$\langle \nabla \hat{F}_p(\mathbf{U}^k) + \frac{1}{\eta}(\mathbf{U}^{k+1} - \mathbf{U}^k), \mathbf{U} - \mathbf{U}^{k+1} \rangle \geq 0. \quad (80)$$

Together with $\mathbf{U}^k \in \mathcal{C}$, we obtain

$$\langle \nabla \hat{F}_p(\mathbf{U}^k), \mathbf{U}^{k+1} - \mathbf{U}^k \rangle + \frac{1}{\eta} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F^2 \leq 0. \quad (81)$$

Since $\hat{F}_p(\mathbf{U})$ is locally L_p -smooth for $\mathbf{U} \in \mathcal{C}$ as shown in Lemma 3 with $0 < \eta \leq 1/L_p$, we have

$$\begin{aligned} & \hat{F}_p(\mathbf{U}^{k+1}) \\ & \leq \hat{F}_p(\mathbf{U}^k) + \langle \nabla \hat{F}_p(\mathbf{U}^k), \mathbf{U}^{k+1} - \mathbf{U}^k \rangle + \frac{L_p}{2} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F^2 \\ & \leq \hat{F}_p(\mathbf{U}^k) - \left(\frac{1}{\eta} - \frac{L_p}{2}\right) \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F^2 \\ & \leq \hat{F}_p(\mathbf{U}^k) - \frac{L_p}{2} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F^2. \end{aligned} \quad (82)$$

It is obvious that $\hat{F}_p(\mathbf{U}^k) > 0$ and we know $\{\hat{F}_p(\mathbf{U}^k)\}$ is decreasing from (82), thus there exists some positive value $\hat{F}_p^*$ such that

$$\lim_{k \rightarrow \infty} \hat{F}_p(\mathbf{U}^k) = \hat{F}_p^*. \quad (83)$$

Besides, by summing the inequality (82) for $k = 0, 1, \dots, K-1$, we obtain

$$\begin{aligned} \frac{L_p}{2} \sum_{k=0}^{K-1} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F^2 & \leq \hat{F}_p(\mathbf{U}^0) - \hat{F}_p(\mathbf{U}^K) \\ & \leq \hat{F}_p(\mathbf{U}^0) - \hat{F}_p^* \leq \hat{F}_p(\mathbf{U}^0), \end{aligned} \quad (84)$$

where the last inequality uses $\hat{F}_p^* > 0$, and thus

$$\min_{k \in [K]} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F \leq \sqrt{\frac{2\hat{F}_p(\mathbf{U}^0)}{L_p K}}. \quad (85)$$

APPENDIX B

PROOF OF THEOREM 2

Before giving the result, we first review an important definition and a useful lemma.

Definition 2 (Gradient mapping). *Suppose $\mathcal{Q} \subseteq \mathbb{C}^d$ is closed, convex, and nonempty, and f is differentiable. Given $t > 0$, the gradient mapping $G_t : \mathbb{C}^d \rightarrow \mathbb{C}^d$ is defined by*

$$G_t(\mathbf{q}) = t(\mathbf{q} - \mathcal{P}_{\mathcal{Q}}(\mathbf{q} - \frac{1}{t} \nabla f(\mathbf{q}))), \quad \mathbf{q} \in \mathbb{R}^d, \quad (86)$$

where $\mathcal{P}_{\mathcal{Q}}(\cdot)$ denotes the Euclidean projection operator.

Lemma 4 ([32]). *Denote $\bar{\mathbf{q}} = \mathcal{P}_{\mathcal{Q}}(\mathbf{q} - \eta \nabla f(\mathbf{q}))$, so that $G_{1/\eta}(\mathbf{q}) = \frac{1}{\eta}(\mathbf{q} - \bar{\mathbf{q}})$. If $\|G_{1/\eta}(\mathbf{q})\| \leq \varepsilon$ for some $\varepsilon \geq 0$, then*

$$-\nabla f(\bar{\mathbf{q}}) \in N_{\mathcal{Q}}(\bar{\mathbf{q}}) + \mathcal{B}_2(0, \varepsilon(L\eta + 1)), \quad (87)$$

where L is the smoothness parameter of $f(\cdot)$.

Then we begin our proof. Define

$$G_{1/\eta}(\mathbf{U}^k) = \frac{1}{\eta}(\mathbf{U}^k - \mathcal{P}_{\mathcal{C}}(\mathbf{U}^k - \eta \nabla \hat{F}_p(\mathbf{U}^k))), \quad (88)$$

which means

$$G_{1/\eta}(\mathbf{U}^k) = \frac{1}{\eta}(\mathbf{U}^k - \mathbf{U}^{k+1}). \quad (89)$$

Since $1/\eta = L_p$, together with Theorem 1, we obtain

$$\min_{k \in [K]} \|G_{1/\eta}(\mathbf{U}^k)\|_F \leq \sqrt{\frac{2L_p \hat{F}_p(\mathbf{U}^0)}{K}}. \quad (90)$$

Since

$$K \geq \frac{8L\hat{F}_p(\mathbf{U}^0)}{\varepsilon^2}, \quad (91)$$

we get

$$\min_{k \in [K]} \|G_{1/\eta}(\mathbf{U}^k)\|_F \leq \frac{\varepsilon}{2}. \quad (92)$$

Together with Lemma 4, we complete the proof.

APPENDIX C
PROOF OF THEOREM 3

Note that $\mathbf{U}^{k+1}$ is the solution to the following optimization problem:

$$\begin{aligned}\mathbf{U}^{k+1} &= \arg \min_{\mathbf{U} \in \mathcal{C}} \langle \nabla \hat{F}_p(\mathbf{U}^k), \mathbf{U} - \tilde{\mathbf{U}}^k \rangle + \frac{1}{2\eta} \|\mathbf{U} - \tilde{\mathbf{U}}^k\|_F^2 \\ &= \mathcal{P}_{\mathcal{C}}(\tilde{\mathbf{U}}^k - \eta \nabla \hat{F}_p(\tilde{\mathbf{U}}^k)),\end{aligned}\quad (93)$$

Then we will have the following descent result:

$$\begin{aligned}\hat{F}_p(\mathbf{U}^{k+1}) &\leq \hat{F}_p(\tilde{\mathbf{U}}^k) + \langle \nabla \hat{F}_p(\tilde{\mathbf{U}}^k), \mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k \rangle + \frac{L_p}{2} \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2 \\ &\leq \hat{F}_p(\tilde{\mathbf{U}}^k) - \left(\frac{1}{\eta} - \frac{L_p}{2}\right) \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2 \\ &\leq \hat{F}_p(\tilde{\mathbf{U}}^k) - \frac{L_p}{2} \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2.\end{aligned}\quad (94)$$

The update rule of our algorithm guarantees that:

$$\hat{F}_p(\tilde{\mathbf{U}}^{k+1}) \leq \hat{F}_p(\mathbf{U}^{k+1}) \leq \hat{F}_p(\tilde{\mathbf{U}}^k) \leq \hat{F}_p(\mathbf{U}^k). \quad (95)$$

As $\hat{F}_p(\tilde{\mathbf{U}}^k), \hat{F}_p(\mathbf{U}^k) > 0$, we can conclude that

$$\lim_{k \rightarrow \infty} \hat{F}_p(\mathbf{U}^k) = \lim_{k \rightarrow \infty} \hat{F}_p(\tilde{\mathbf{U}}^k) = \hat{F}_p^\diamond, \quad (96)$$

From (95), by induction we conclude that for all k

$$\hat{F}_p(\tilde{\mathbf{U}}^k) \leq \hat{F}_p(\mathbf{U}^0), \quad \hat{F}_p(\mathbf{U}^k) \leq \hat{F}_p(\mathbf{U}^0).$$

As $\mathbf{U}^k \in \mathcal{C}$, we have $\|\mathbf{U}^k\|_F \leq \sqrt{n} \|\mathbf{U}^k\|_{2,\infty} \leq \frac{2}{1+2\beta} \sqrt{\mu r \sigma}$. Besides,

$$\begin{aligned}\|\tilde{\mathbf{U}}^{k+1}\|_F &= \|(1+\beta)\mathbf{U}^{k+1} - \beta\mathbf{U}^k\|_F \\ &\leq (1+\beta)\|\mathbf{U}^{k+1}\|_F + \beta\|\mathbf{U}^k\|_F \leq 2\sqrt{\mu r \sigma},\end{aligned}$$

where $\mathbf{U}^{k+1}, \mathbf{U}^k \in \mathcal{C}'$. Given $\tilde{\mathbf{U}}^0 \in \mathcal{C}'$, we can conclude that $\{\tilde{\mathbf{U}}^k\}, \{\mathbf{U}^k\}$ are bounded and thus we have bounded limit points. Combining (94) and (95) yields

$$\begin{aligned}\frac{L_p}{2} \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2 &\leq \hat{F}_p(\tilde{\mathbf{U}}^k) - \hat{F}_p(\mathbf{U}^{k+1}) \\ &\leq \hat{F}_p(\tilde{\mathbf{U}}^k) - \hat{F}_p(\tilde{\mathbf{U}}^{k+1}).\end{aligned}$$

By telescoping over k and letting $k \rightarrow \infty$, we obtain that

$$\frac{L_p}{2} \sum_{k=1}^{\infty} \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2 \leq \hat{F}_p(\tilde{\mathbf{U}}^1) - \inf \hat{F}_p < \hat{F}_p(\tilde{\mathbf{U}}^1) < \infty,$$

and this implies that $\|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F \rightarrow 0$ and hence that $\mathbf{U}^{k+1}$ and $\tilde{\mathbf{U}}^k$ share the same set of limit points Ω . As Ω is closed and bounded, we conclude that Ω is compact in $\mathbb{R}^d$.

Consider any limit point $\mathbf{U}' \in \Omega$, and w.l.o.g. we write that $\mathbf{U}^{k+1} \rightarrow \mathbf{U}'$, $\tilde{\mathbf{U}}^k \rightarrow \mathbf{U}'$ by restricting to a subsequence. By the optimality condition of the gradient updating step in (93), we have

$$-(\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k) - \eta \nabla \hat{F}_p(\tilde{\mathbf{U}}^k) \in N_{\mathcal{C}}(\mathbf{U}^{k+1}).$$

Noting that $\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k \rightarrow \mathbf{0}$ and $\mathbf{U}^{k+1}, \tilde{\mathbf{U}}^k \rightarrow \mathbf{U}'$, we conclude that $-\nabla \hat{F}_p(\mathbf{U}') \in N_{\mathcal{C}}(\mathbf{U}')$. By (96), we know $\hat{F}_p(\mathbf{U}') = \hat{F}_p^\diamond$ for any $\mathbf{U}' \in \Omega$.

APPENDIX D
PROOF OF THEOREM 4

From (93), we know

$$-\nabla \hat{F}_p(\tilde{\mathbf{U}}^k) - \frac{1}{\eta}(\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k) \in N_{\mathcal{C}}(\mathbf{U}^{k+1}),$$

and it further implies that

$$\begin{aligned}\nabla \hat{F}_p(\mathbf{U}^{k+1}) - \nabla \hat{F}_p(\tilde{\mathbf{U}}^k) - \frac{1}{\eta}(\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k) \\ \in \nabla \hat{F}_p(\mathbf{U}^{k+1}) + N_{\mathcal{C}}(\mathbf{U}^{k+1}).\end{aligned}$$

Let $\mathbf{E} = \nabla \hat{F}_p(\mathbf{U}^{k+1}) - \nabla \hat{F}_p(\tilde{\mathbf{U}}^k) - \frac{1}{\eta}(\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k)$. For $\tilde{\mathbf{U}}^k \in \mathcal{C}' \subseteq \mathcal{C}$, it is easy to verify that $\mathbf{U}^{k+1} \in \mathcal{C}$. As $\mathbf{U}^{k+1}, \tilde{\mathbf{U}}^k \in \mathcal{C}$ and $\hat{F}_p(\cdot)$ is locally L_p -Lipschitz from Lemma 3, we can obtain

$$\begin{aligned}\|\mathbf{E}\|_F &\leq \|\nabla \hat{F}_p(\mathbf{U}^{k+1}) - \nabla \hat{F}_p(\tilde{\mathbf{U}}^k)\|_F + \frac{1}{\eta} \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F \\ &\leq (L_p + \frac{1}{\eta}) \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F,\end{aligned}$$

We know

$$\partial G(\mathbf{U}^{k+1}) = \nabla \hat{F}_p(\mathbf{U}^{k+1}) + N_{\mathcal{C}}(\mathbf{U}^{k+1}),$$

and we give the form of the distance as

$$\text{dist}_{\partial G(\mathbf{U})}(\mathbf{0}) = \min_{\mathbf{U} \in \partial G(\mathbf{U})} \|\mathbf{U}\|_F.$$

Combining (94) and (95), we know

$$\hat{F}_p(\mathbf{U}^{k+1}) \leq \hat{F}_p(\mathbf{U}^k) - \frac{L_p}{2} \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2.$$

As $\mathbf{E} \in \partial G(\mathbf{U}^{k+1})$, we have

$$\text{dist}_{\partial G(\mathbf{U}^{k+1})}(\mathbf{0}) \leq \|\mathbf{E}\|_F \leq \left(L_p + \frac{1}{\eta}\right) \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F,$$

From the KL property of $G(\mathbf{U}^{k+1})$ and $G(\mathbf{U}^{k+1}) = \hat{F}_p(\mathbf{U}^{k+1})$ as $\mathbf{U}^{k+1} \in \mathcal{C}'$:

$$\varphi'(\hat{F}_p(\mathbf{U}^{k+1}) - \hat{F}_p^\diamond) \text{dist}_{\partial G(\mathbf{U}^{k+1})}(\mathbf{0}) \geq \frac{1}{c_{\text{KL}}},$$

where $\varphi(t) = 2t^{\frac{1}{2}}$. Let $r_{k+1} = \hat{F}_p(\mathbf{U}^{k+1}) - \hat{F}_p^\diamond$, we have

$$\begin{aligned}1 &\leq (c_{\text{KL}} \varphi'(r_{k+1}) \text{dist}_{\partial G(\mathbf{U}^{k+1})}(\mathbf{0}))^2 \\ &\leq c_{\text{KL}}^2 (\varphi'(r_{k+1}))^2 (L_p + \frac{1}{\eta})^2 \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2 \\ &\leq c_{\text{KL}}^2 (\varphi'(r_{k+1}))^2 \frac{2(L_p + \frac{1}{\eta})^2}{L_p} (\hat{F}_p(\mathbf{U}^k) - \hat{F}_p(\mathbf{U}^{k+1})) \\ &\leq c_{\text{KL}}^2 \frac{2(L_p + \frac{1}{\eta})^2}{L_p} (r_k - r_{k+1}) (\varphi'(r_{k+1}))^2,\end{aligned}$$

and inserting $\varphi'(t) = \frac{1}{\sqrt{t}}$ and $\eta = \frac{1}{L_p}$, we establish that

$$r_{k+1} \leq \left(1 - \frac{1}{1 + C_p}\right) r_k,$$

where $C_p = 8c_{\text{KL}}^2 L_p$. Finally, we can prove Theorem 4 via the induction step.

APPENDIX E
PROOF OF LEMMA 1

When $p = 1$, we get

$$\hat{f}_p(\mathbf{U}) = \|\phi_b(\mathbf{w})\|_1 = \langle \mathbf{1}, \phi_b(\mathbf{w}) \rangle, \quad (97)$$

then the differential of $\hat{f}_p(\mathbf{U})$ is

$$\begin{aligned} d\hat{f}_p(\mathbf{U}) &= \langle \mathbf{1}, \phi'_b(\mathbf{w}) \odot d\mathbf{w} \rangle = \langle \phi'_b(\mathbf{w}), d\mathbf{w} \rangle \\ &= \frac{1}{2} \langle \phi'_b(\mathbf{w}), \bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\mathbf{U}\mathbf{U}^T + \mathbf{U}d\mathbf{U}^T) \rangle \\ &\quad + \frac{1}{2} \langle \phi'_b(\mathbf{w}), \mathbf{y} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\bar{\mathbf{U}}\mathbf{U}^H + \bar{\mathbf{U}}d\mathbf{U}^H) \rangle \\ &= \langle \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y}), d\mathbf{U}\mathbf{U}^T \rangle \\ &\quad + \langle \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}}), d\bar{\mathbf{U}}\mathbf{U}^H \rangle \\ &= \langle \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{\mathbf{U}}, d\mathbf{U} \rangle \\ &\quad + \langle \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}})\mathbf{U}, d\bar{\mathbf{U}} \rangle \end{aligned} \quad (98)$$

where the second and fourth equalities apply

$$\langle \mathbf{A}, \mathbf{B} \odot \mathbf{D} \rangle = \langle \mathbf{A} \odot \bar{\mathbf{B}}, \mathbf{D} \rangle, \quad \mathbf{A}, \mathbf{B}, \mathbf{D} \in \mathbb{C}^{m \times n}. \quad (99)$$

and the third equality uses

$$\begin{aligned} d\mathbf{w} &= \frac{1}{2} \bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\mathbf{U}\mathbf{U}^T + \mathbf{U}d\mathbf{U}^T) \\ &\quad + \frac{1}{2} \mathbf{y} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\bar{\mathbf{U}}\mathbf{U}^H + \bar{\mathbf{U}}d\mathbf{U}^H). \end{aligned} \quad (100)$$

Therefore, we get

$$\nabla \hat{f}_p(\mathbf{U}) = \nabla_{\bar{\mathbf{U}}} \hat{f}_p(\mathbf{U}) = \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{\mathbf{U}}. \quad (101)$$

When $p = 2$, using Facts 1 and 2 yields

$$\hat{f}_p(\mathbf{U}) = \frac{1}{2} \|\phi_b(\mathbf{w})\|_2^2 = \frac{1}{2} \langle \phi_b(\mathbf{w}), \phi_b(\mathbf{w}) \rangle, \quad (102)$$

then the differential of $\hat{f}_p(\mathbf{U})$ is

$$\begin{aligned} d\hat{f}_p(\mathbf{U}) &= \frac{1}{2} \langle \phi_b(\mathbf{w}), \phi'_b(\mathbf{w}) \odot d\mathbf{w} \rangle + \frac{1}{2} \langle \phi'_b(\mathbf{w}) \odot d\mathbf{w}, \phi_b(\mathbf{w}) \rangle \\ &= \langle \phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}), d\mathbf{w} \rangle \\ &= \frac{1}{2} \langle \phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}), \bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\mathbf{U}\mathbf{U}^T + \mathbf{U}d\mathbf{U}^T) \rangle \\ &\quad + \frac{1}{2} \langle \phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}), \mathbf{y} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\bar{\mathbf{U}}\mathbf{U}^H + \bar{\mathbf{U}}d\mathbf{U}^H) \rangle \\ &= \langle \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y}), d\mathbf{U}\mathbf{U}^T \rangle \\ &\quad + \langle \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}}), d\bar{\mathbf{U}}\mathbf{U}^H \rangle \\ &= \langle \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{\mathbf{U}}, d\mathbf{U} \rangle \\ &\quad + \langle \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}})\mathbf{U}, d\bar{\mathbf{U}} \rangle, \end{aligned} \quad (103)$$

where the second equality applies $d\mathbf{w} = d\bar{\mathbf{w}}$, the third equality uses (100) and the fourth equality applies (99). Therefore, using Facts 1 and 2 obtains

$$\nabla \hat{f}_p(\mathbf{U}) = \nabla_{\bar{\mathbf{U}}} \hat{f}_p(\mathbf{U}) = \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{\mathbf{U}}. \quad (104)$$

APPENDIX F
PROOF OF LEMMA 3

From Lemma 2, using the triangle inequality yields

$$\|\nabla^2 \hat{F}_p(\mathbf{U})\| \leq 2(\|\mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{F}_p}\| + \|\mathbf{H}_{\mathbf{U}, \bar{\mathbf{U}}}^{\hat{F}_p}\|). \quad (105)$$

Before giving the upper bound of $\|\nabla^2 \hat{F}_p(\mathbf{U})\|$, we first provide the upper bounds for other important components.

In terms of $\|\mathbf{\Gamma}_g\|$ and $\|\mathbf{\Gamma}_f\|$, noticing that $\sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T$ and $\mathbf{I}_{s^2} - \sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T$ are projection matrices, we have

$$\left\| \mathbf{I}_{s^2} - \sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T \right\| \leq 1, \quad (106)$$

$$\left\| \sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T \right\| \leq 1. \quad (107)$$

Together with the fact that $\|\mathbf{K}_{s^2}\| = 1$, we have

$$\begin{aligned} \|\mathbf{\Gamma}_g\| &\leq (\|\mathbf{I}_{s^2}\| + \|\mathbf{K}_{s^2}\|) \left\| \mathbf{I}_{s^2} - \sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T \right\| \\ &\leq 2, \end{aligned} \quad (108)$$

$$\|\mathbf{\Gamma}_f\| \leq (\|\mathbf{I}_{s^2}\| + \|\mathbf{K}_{s^2}\|) \left\| \sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T \right\| \leq 2. \quad (109)$$

As for $\|\mathbf{U}\|$, we obtain

$$\|\mathbf{U}\| \leq \|\mathbf{U}\|_{\text{F}} \leq \sqrt{n} \|\mathbf{U}\|_{2, \infty} \leq 2\sqrt{\mu r \sigma}. \quad (110)$$

Using the definition of $\phi_b(\cdot)$, we obtain

$$\|\phi'_b(\mathbf{w})\|_{\infty} \leq 1, \quad (111)$$

$$\|\phi''_b(\mathbf{w})\|_{\infty} \leq \frac{1}{2\sqrt{b}}, \quad (112)$$

$$\|\phi_b(\mathbf{w}) \odot \phi''_b(\mathbf{w})\|_{\infty} \leq 1/2, \quad (113)$$

$$\|\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w})\|_{\infty} \leq \sqrt{\frac{8\mu^2 r^2 \sigma^2}{n^2} + \frac{b}{4}} + \frac{2\sqrt{2}\mu r \sigma}{n}, \quad (114)$$

where the last line applies the upper bound of $\|\mathbf{w}\|_{\infty}$ that

$$\begin{aligned} \|\mathbf{w}\|_{\infty} &= \|\text{Re}(\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T))\|_{\infty} \\ &= \|\bar{\mathbf{y}}\|_{\infty} \|\mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)\|_{\infty} \\ &\leq \sqrt{2} \|\mathbf{U}\mathbf{U}^T\|_{\infty} \leq \sqrt{2} \|\mathbf{U}\|_{2, \infty}^2 \leq \frac{4\sqrt{2}\mu r \sigma}{n}, \end{aligned}$$

Together with $\|\mathcal{G}\mathcal{D}^{-1}\| \leq 1$ and $\|\mathbf{y}\| = \sqrt{2n}$, we have

$$\|\mathcal{G}\mathcal{D}^{-1}(\phi''_b(\mathbf{w}))\|_{\infty} \leq \|\phi''_b(\mathbf{w})\|_{\infty} \leq \frac{1}{2\sqrt{b}}, \quad (115)$$

$$\|\mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y})\| \leq \|\phi'_b(\mathbf{w})\|_{\infty} \|\mathbf{y}\| \leq \sqrt{2n}, \quad (116)$$

$$\|\mathcal{G}\mathcal{D}^{-1}(\phi''_b(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y})\|_{\infty} \leq \|\phi''_b(\mathbf{w})\|_{\infty} \|\mathbf{y}\|_{\infty}^2 \leq \frac{1}{\sqrt{b}}, \quad (117)$$

$$\|\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y}\| \leq \|\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w})\|_{\infty} \|\mathbf{y}\| \quad (118)$$

When $p = 1$, we establish the upper bounds as follows:

$$\|\mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{F}_1}\| \leq \|\mathbf{\Gamma}_f\| \|\mathbf{U}\|^2 \|\mathcal{G}\mathcal{D}^{-1}(\phi''_b(\mathbf{w}))\|_{\infty} + \gamma \|\mathbf{\Gamma}_g\| \|\mathbf{U}\|^2,$$

$$\leq 4\mu\sigma\left(\frac{1}{\sqrt{b}} + 2\gamma\right), \quad (119)$$

and

$$\begin{aligned} \|\mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{\mathbf{F}}_1}\| &\leq \|\mathcal{GD}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y})\| \\ &\quad + \frac{1}{2}\|\mathbf{I}_f\|\|\mathbf{U}\|^2\|\mathcal{GD}^{-1}(\phi''_b(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y})\|_\infty + \gamma\|\mathbf{U}\|^2 \\ &\leq \sqrt{2n} + 4\mu\sigma\left(\frac{1}{\sqrt{b}} + \gamma\right). \end{aligned} \quad (120)$$

Consequently, when $p = 1$, we get

$$L_p = 2\sqrt{2n} + 8\mu\sigma\left(\frac{2}{\sqrt{b}} + 3\gamma\right). \quad (121)$$

Similarly, when $p = 2$, we establish the upper bounds as follows after a careful calculation:

$$\begin{aligned} \|\mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{\mathbf{F}}_2}\| &\leq \|\mathbf{I}_f\|\|\mathbf{U}\|^2\|\mathcal{GD}^{-1}(\phi'_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) + \phi_b(\mathbf{w}) \odot \phi''_b(\mathbf{w}))\|_\infty \\ &\quad + \gamma\|\mathbf{I}_g\|\|\mathbf{U}\|^2 \\ &\leq 4\mu\sigma(3 + 2\gamma), \end{aligned} \quad (122)$$

and

$$\begin{aligned} \|\mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{\mathbf{F}}_2}\| &\leq \|\mathcal{GD}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y})\| + \gamma\|\mathbf{U}\|^2 \\ &\quad + \frac{1}{2}\|\mathbf{I}_f\|\|\mathbf{U}\|^2\|\mathcal{GD}^{-1}(\Lambda_\phi(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y})\|_\infty \\ &\leq (\sqrt{16\mu^2r^2\sigma^2/n + nb/2} + 4\mu\sigma/\sqrt{n}) + \frac{\mu\sigma(3 + \gamma)}{2} \\ &\leq 4(\sqrt{\mu^2r^2\sigma^2/n + nb} + \mu\sigma/\sqrt{n}) + \frac{\mu\sigma(3 + \gamma)}{2}. \end{aligned}$$

Consequently, when $p = 2$,

$$L_p = 8(\sqrt{\mu^2r^2\sigma^2/n + nb} + \mu\sigma/\sqrt{n}) + \mu\sigma(9 + 3\gamma). \quad (123)$$

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Supplementary Material

APPENDIX G

NUMERICAL VERIFICATION OF KL PROPERTY FOR (73)

A key property of the KL constant is that a smaller $1/c_{\text{KL}}$ in (73) corresponds to a slower convergence rate of our algorithms. Therefore, we seek to verify whether our algorithms' $1/c_{\text{KL}}$ is as large as possible—a feature that directly indicates stable and efficient convergence. We present a plot of $\varphi'(G(\mathbf{U}^k) - \hat{F}_p^\diamond) \text{dist}_{\partial G(\mathbf{U}^k)}(\mathbf{0})$ corresponding to the left part in (73) to demonstrate the algorithms have a relatively large value of $1/c_{\text{KL}}$.

In the first experiment, we test $\varphi'(G(\mathbf{U}^k) - \hat{F}_p^\diamond) \text{dist}_{\partial G(\mathbf{U}^k)}(\mathbf{0})$ of the algorithm based on the ℓ_1 norm in the loss function, versus different iterations under different signal lengths $n = \{32, 40, 48, 56, 64\}$ and $\text{SNR} = \{-10, 0, 10, 20, 30, 40\}$ dB. In the second experiment, we apply the ℓ_2 norm in the loss function and test the algorithm. Other settings are the same as the first experiment.

From Fig. 11 and Fig. 12, we observe that $\varphi'(G(\mathbf{U}^k) - \hat{F}_p^\diamond) \text{dist}_{\partial G(\mathbf{U}^k)}(\mathbf{0})$ is lower bounded and thus there exists a constant $1/c_{\text{KL}}$ in (73) as large as possible to match this lower bound.

Therefore, we finally verify that there exists a relatively large and stable KL constant $1/c_{\text{KL}}$ in our accelerated algorithms such that they exhibit stable and fast linear convergence both theoretically and empirically.

APPENDIX H

EFFECT OF RANK MIS-SPECIFICATION

The proposed formulation assumes that the lifting Hankel matrix $\mathcal{H}(z)$ is exactly of rank r , which corresponds to the standard spectral model where the signal consists of r complex exponentials. In practice, however, the true model order may be unknown. This section examines the behavior of the proposed algorithm under rank mis-specification.

Fig. 13 illustrates the recovered angle-amplitude spectra obtained by Acc-OSCAR- ℓ_1 and Acc-OSCAR- ℓ_2 when the assumed rank $\hat{r}$ differs from the true rank $r_{\text{true}} = 2$ under $\theta = [20.3^\circ, 24.5^\circ]$ and $n = 63$.

When the assumed rank exceeds the true rank ($\hat{r} > r$), the feasible set strictly contains the ground-truth solution. Empirically, mild overestimation introduces additional spurious components, while the other spectral components remain well aligned with the ground truth. Such robustness to rank overestimation is consistent with observations in low-rank matrix factorization and implicit regularization theory.

If the assumed rank is smaller than the true rank ($\hat{r} < r$), the model becomes misspecified and the true Hankel matrix is no longer feasible. The algorithm then recovers the best rank- $\hat{r}$ approximation consistent with the one-bit measurements. As shown in Fig. 13, this leads to missing spectral components and degraded angle estimation. This limitation is inherent to rank-constrained formulations and is not specific to the proposed method.

Both theoretical considerations and numerical evidence suggest that the proposed Acc-OSCAR algorithms are more tolerant to mild rank overestimation than to rank underestimation.

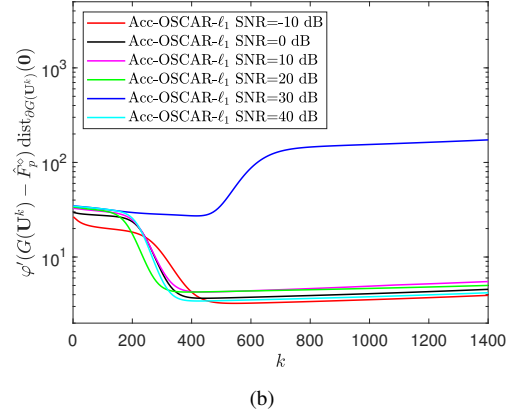
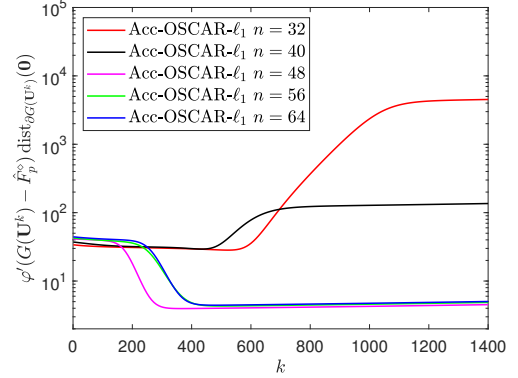


Fig. 11. KL constant vs k for Acc-OSCAR- ℓ_1 . (a) Different cases of $n = \{32, 40, 48, 56, 64\}$. (b) Different cases of $\text{SNR} = \{-10, 0, 10, 20, 30, 40\}$ dB.

When the model order is unknown, slightly overestimating r is therefore preferable, which aligns with common practice in subspace-based spectral estimation methods.

APPENDIX I

EFFECT OF THE SMOOTHING PARAMETER

The smoothing parameter b controls the approximation quality of the one-sided function $[x]_-$ and directly affects both the optimization landscape and the smoothness constants in the convergence analysis.

To examine its practical impact, Fig. 14 reports the performance of Acc-OSCAR- ℓ_1 under different values of b with $n = 255$, $r = 3$ and $\theta = [-56.3^\circ, -15.4^\circ, 30.5^\circ]$. When b is very small, the smoothed objective becomes sharply curved, which may lead to unstable optimization behavior, especially at low SNR. On the other hand, excessively large b introduces noticeable smoothing bias and degrades estimation accuracy. In contrast, moderate values of b provide a favorable trade-off between numerical stability and recovery accuracy, yielding consistently high success rates and low RMSE across a wide range of SNRs.

Based on both empirical evidence and its role in the theoretical smoothness constant L_p , selecting b in a moderate range (e.g., 10^{-4} to 10^{-3} in our experiments) provides robust performance without requiring delicate tuning.

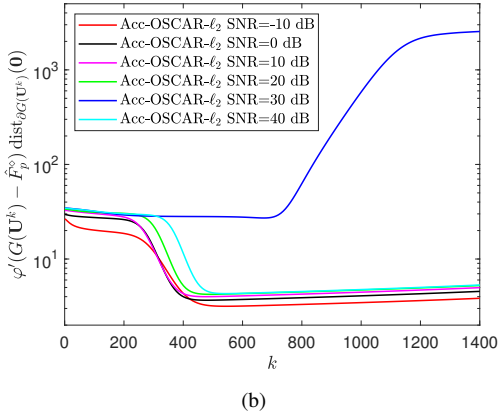
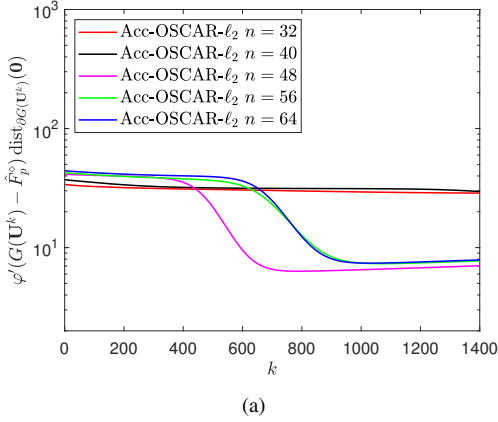


Fig. 12. KL constant vs k for Acc-OSCAR- ℓ_2 . (a) Different cases of $n = \{32, 40, 48, 56, 64\}$. (b) Different cases of $\text{SNR} = \{-10, 0, 10, 20, 30, 40\}$ dB.

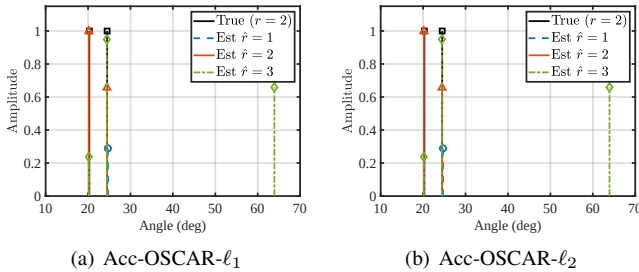


Fig. 13. Angle-amplitude spectra obtained by Acc-OSCAR- ℓ_1 and Acc-OSCAR- ℓ_2 under different assumed ranks. The true signal contains $r_{\text{true}} = 2$ components. Estimated spectra are shown for $\hat{r} = 1, 2, 3$. Underestimation ($\hat{r} = 1$) leads to missing components. Correct specification ($\hat{r} = 2$) yields accurate recovery. Overestimation ($\hat{r} = 3$) introduces an additional spurious component while preserving the true angles.

APPENDIX J PROOF OF LEMMA 2

Before proving Lemma 2, we first give a useful lemma.

Lemma 5. For any $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$, we have

$$\mathcal{G}(\mathbf{u}) \odot \mathcal{G}(\mathbf{v}) = \mathcal{G}\mathcal{D}^{-1}(\mathbf{u} \odot \mathbf{v}). \quad (124)$$

Proof. According to the definition of $\mathcal{H}$, we have

$$\mathcal{H}(\mathbf{u}') \odot \mathcal{H}(\mathbf{v}') = \mathcal{H}(\mathbf{u}' \odot \mathbf{v}'), \quad \mathbf{u}', \mathbf{v}' \in \mathbb{C}^n. \quad (125)$$

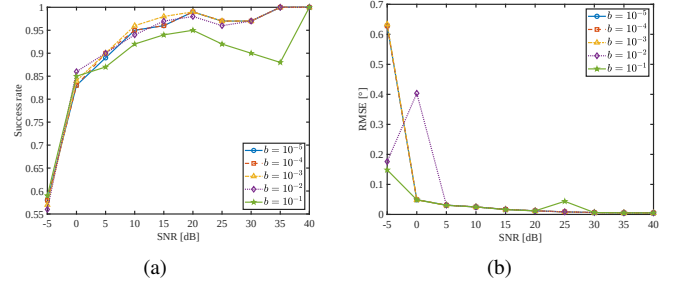


Fig. 14. Performance of Acc-OSCAR- ℓ_1 under different smoothing parameters b with $n = 255$ and $r = 3$. (a) Success rate vs. SNR. (b) RMSE vs. SNR.

Define $\mathbf{u} = \mathcal{D}\mathbf{u}'$ and $\mathbf{v} = \mathcal{D}\mathbf{v}'$. We obtain

$$\mathcal{H}\mathcal{D}^{-1}(\mathbf{u}) \odot \mathcal{H}\mathcal{D}^{-1}(\mathbf{v}) = \mathcal{H}(\mathcal{D}^{-1}\mathbf{u} \odot \mathcal{D}^{-1}\mathbf{v}) \quad (126)$$

$$= \mathcal{H}\mathcal{D}^{-2}(\mathbf{u} \odot \mathbf{v}). \quad (127)$$

Using the fact $\mathcal{G} = \mathcal{H}\mathcal{D}^{-1}$ completes the proof. $\square$

Then we start to prove Lemma 2. Based on (27), the Hessian matrices $\nabla^2 \hat{F}_p(\mathbf{U})$ are

$$\nabla^2 \hat{F}_p(\mathbf{U}) = \nabla^2 \hat{f}_p(\mathbf{U}) + \gamma \nabla^2 g(\mathbf{U}), \quad p = 1, 2. \quad (128)$$

Here, $\nabla^2 \hat{f}_p(\mathbf{U})$ and $\nabla^2 g(\mathbf{U})$ are the Hessian matrices of $\hat{f}_p(\mathbf{U})$ and $g(\mathbf{U})$, defined by

$$\nabla^2 \hat{f}_p(\mathbf{U}) = \begin{bmatrix} \mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{f}_p} & \mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{f}_p} \\ \mathbf{H}_{\mathbf{U}, \mathbf{U}}^{\hat{f}_p} & \mathbf{H}_{\mathbf{U}, \bar{\mathbf{U}}}^{\hat{f}_p} \end{bmatrix}, \quad (129)$$

$$\nabla^2 g(\mathbf{U}) = \begin{bmatrix} \mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^g & \mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^g \\ \mathbf{H}_{\mathbf{U}, \mathbf{U}}^g & \mathbf{H}_{\mathbf{U}, \bar{\mathbf{U}}}^g \end{bmatrix}, \quad (130)$$

where $\mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{f}_p}$, $\mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{f}_p}$, $\mathbf{H}_{\mathbf{U}, \mathbf{U}}^{\hat{f}_p}$ and $\mathbf{H}_{\mathbf{U}, \bar{\mathbf{U}}}^{\hat{f}_p}$ are the complex Hessian matrices of $\hat{f}_p(\mathbf{U})$ and $\mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^g$, $\mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^g$, $\mathbf{H}_{\mathbf{U}, \mathbf{U}}^g$ and $\mathbf{H}_{\mathbf{U}, \bar{\mathbf{U}}}^g$ are the complex Hessian matrices of $g(\mathbf{U})$ in Fact 3.

Since $\nabla^2 g(\mathbf{U})$ appears in both $p = 1$ and $p = 2$, we consider $\nabla^2 g(\mathbf{U})$ first. As shown in (37), we have provided the gradient $\nabla g(\mathbf{U})$. The differential of $\nabla_{\bar{\mathbf{U}}} g(\mathbf{U}) = \nabla g(\mathbf{U})$ is

$$\begin{aligned} d\nabla_{\bar{\mathbf{U}}} g(\mathbf{U}) &= (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(d\mathbf{U}\mathbf{U}^T)\bar{\mathbf{U}} + (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}d\mathbf{U}^T)\bar{\mathbf{U}} \\ &\quad + (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T)d\bar{\mathbf{U}}. \end{aligned} \quad (131)$$

Define

$$\mathbf{M}_1 = (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(d\mathbf{U}\mathbf{U}^T)\bar{\mathbf{U}}, \quad (132)$$

$$\mathbf{M}_2 = (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}d\mathbf{U}^T)\bar{\mathbf{U}}, \quad (133)$$

$$\mathbf{M}_3 = (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T)d\bar{\mathbf{U}}. \quad (134)$$

Then using (49), we have

$$\mathbf{M}_1 = d\mathbf{U}\mathbf{U}^T\bar{\mathbf{U}} - \sum_{a=1}^n \langle \mathbf{H}_a, d\mathbf{U}\mathbf{U}^T \rangle \mathbf{H}_a \bar{\mathbf{U}}, \quad (135)$$

$$\mathbf{M}_2 = \mathbf{U}d\mathbf{U}^T\bar{\mathbf{U}} - \sum_{a=1}^n \langle \mathbf{H}_a, \mathbf{U}d\mathbf{U}^T \rangle \mathbf{H}_a \bar{\mathbf{U}} \quad (136)$$

Recalling the property of the Kronecker product, for $A \in \mathbb{C}^{n_1 \times n_2}$, $X \in \mathbb{C}^{n_2 \times n_3}$, $B \in \mathbb{C}^{n_3 \times n_4}$, we know

$$\text{vec}(AXB) = (B^T \otimes A)\text{vec}(X). \quad (137)$$

Then for M_1 , we have

$$\begin{aligned} \text{vec}(M_1) &= (U^H U \otimes I_s) \text{vec}(dU) - (U^H \otimes I_s) \\ &\quad \cdot \left[\sum_{a=1}^n \text{vec}(H_a) \text{vec}(H_a)^T \right] \text{vec}(dU U^T) \\ &= (U^H U \otimes I_s) \text{vec}(dU) - (U^H \otimes I_s) \\ &\quad \cdot \left[\sum_{a=1}^n \text{vec}(H_a) \text{vec}(H_a)^T \right] (U \otimes I_s) \text{vec}(dU). \end{aligned} \quad (138)$$

For M_2 , we have

$$\begin{aligned} \text{vec}(M_2) &= (U^H \otimes I_s) \text{vec}(U dU^T) - (U^H \otimes I_s) \\ &\quad \cdot \left[\sum_{a=1}^n \text{vec}(H_a) \text{vec}(H_a)^T \right] \text{vec}(U dU^T) \\ &= (U^H \otimes I_s) K_{s^2} (U \otimes I_s) \text{vec}(dU) - (U^H \otimes I_s) \\ &\quad \cdot \left[\sum_{a=1}^n \text{vec}(H_a) \text{vec}(H_a)^T \right] K_{s^2} (U \otimes I_s) \text{vec}(dU). \end{aligned} \quad (139)$$

For M_3 , we have

$$\text{vec}(M_3) = (I_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UU^T)) \text{vec}(d\bar{U}). \quad (140)$$

Therefore,

$$\begin{aligned} \text{vec}(d\nabla g(U)) &= (U^H \otimes I_s) \left(I_{s^2} - \sum_{a=1}^n \text{vec}(H_a) \text{vec}(H_a)^T \right) \\ &\quad \cdot (I_{s^2} + K_{s^2}) (U \otimes I_s) \text{vec}(dU) \\ &\quad + (I_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UU^T)) \text{vec}(d\bar{U}) \\ &= (U^H \otimes I_s) \Gamma_g (U \otimes I_s) \text{vec}(dU) \\ &\quad + (I_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UU^T)) \text{vec}(d\bar{U}). \end{aligned} \quad (141)$$

As for $\nabla_U g(U) = \overline{\nabla_{\bar{U}} g(\bar{U})}$, we have

$$\begin{aligned} \text{vec}(d\nabla_U g(U)) &= \overline{\text{vec}(d\nabla_{\bar{U}} g(\bar{U}))} \\ &= (U^T \otimes I_s) \Gamma_g (\bar{U} \otimes I_s) \text{vec}(d\bar{U}) \\ &\quad + (I_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\bar{U}U^H)) \text{vec}(dU). \end{aligned} \quad (142)$$

Then we obtain

$$\begin{aligned} d^2 g(U) &= \left(\text{vec}(d\nabla_U g(U)) \right)^T \text{vec}(dU) \\ &\quad + \left(\text{vec}(d\nabla_{\bar{U}} g(\bar{U})) \right)^T \text{vec}(d\bar{U}). \end{aligned} \quad (143)$$

Combining (141), (142) and (143) gives

$$H_{\bar{U},U}^g = (U^T \otimes I_s) \Gamma_g (\bar{U} \otimes I_s), \quad (144)$$

$$H_{U,\bar{U}}^g = (U^H \otimes I_s) \Gamma_g (U \otimes I_s), \quad (145)$$

$$H_{U,U}^g = I_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\bar{U}U^H), \quad (146)$$

$$H_{\bar{U},\bar{U}}^g = I_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UU^T). \quad (147)$$

Next, we consider the Hessian matrices of $\hat{f}_p(U)$ for $p = 1$ and $p = 2$. When $p = 1$, according to Eq. (98), we get

$$\nabla_{\bar{U}} \hat{f}_1(U) = \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y}) \bar{U}, \quad (148)$$

$$\nabla_U \hat{f}_1(U) = \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}}) U. \quad (149)$$

Together with Lemma 5, after careful calculation, we obtain

$$H_{\bar{U},U}^{\hat{f}_1} = (U^T \otimes I_s) \Gamma_f \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\phi''_b(\mathbf{w}))) (\bar{U} \otimes I_s), \quad (150)$$

$$H_{U,\bar{U}}^{\hat{f}_1} = (U^H \otimes I_s) \Gamma_f \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\phi''_b(\mathbf{w}))) (U \otimes I_s), \quad (151)$$

$$\begin{aligned} H_{U,U}^{\hat{f}_1} &= I_r \otimes \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}}) + \frac{1}{2} (U^T \otimes I_s) \Gamma_f \\ &\quad \cdot \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\phi''_b(\mathbf{w}) \odot \bar{\mathbf{y}} \odot \bar{\mathbf{y}})) (U \otimes I_s), \end{aligned} \quad (152)$$

$$\begin{aligned} H_{\bar{U},\bar{U}}^{\hat{f}_1} &= I_r \otimes \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y}) + \frac{1}{2} (U^H \otimes I_s) \Gamma_f \\ &\quad \cdot \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\phi''_b(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y})) (\bar{U} \otimes I_s). \end{aligned} \quad (153)$$

When $p = 2$, we have

$$\nabla_{\bar{U}} \hat{f}_2(U) = \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y}) \bar{U}, \quad (154)$$

$$\nabla_U \hat{f}_2(U) = \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}}) U. \quad (155)$$

After careful calculation, we obtain

$$H_{\bar{U},U}^{\hat{f}_2} = (U^T \otimes I_s) \Gamma_f \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\Lambda_\phi(\mathbf{w}))) (\bar{U} \otimes I_s), \quad (156)$$

$$H_{U,\bar{U}}^{\hat{f}_2} = (U^H \otimes I_s) \Gamma_f \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\Lambda_\phi(\mathbf{w}))) (U \otimes I_s), \quad (157)$$

$$\begin{aligned} H_{U,U}^{\hat{f}_2} &= I_r \otimes \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}}) + \frac{1}{2} (U^T \otimes I_s) \Gamma_f \\ &\quad \cdot \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\Lambda_\phi(\mathbf{w}) \odot \bar{\mathbf{y}} \odot \bar{\mathbf{y}})) (U \otimes I_s), \end{aligned} \quad (158)$$

$$\begin{aligned} H_{\bar{U},\bar{U}}^{\hat{f}_2} &= I_r \otimes \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y}) + \frac{1}{2} (U^H \otimes I_s) \Gamma_f \\ &\quad \cdot \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\Lambda_\phi(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y})) (\bar{U} \otimes I_s). \end{aligned} \quad (159)$$